FedCD: A Fairness-aware Federated Cognitive Diagnosis Framework

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With the rapid development of Internet technologies, online intelligent education platforms have generated a vast amount of distributed student learning data. This influx of data presents opportunities for cognitive diagnosis (CD) to assess students' mastery of knowledge concepts while also raising significant data privacy and security challenges. To cope with this issue, federated learning (FL) becomes a promising solution by jointly training models across multiple local clients without sharing their original data. However, heterogeneous response distributions across different schools, arising from differences in student populations, educational contexts, and learning behaviors, pose an additional challenge to client fairness in federated CD. To address this challenge, this paper proposes a fairness-aware **fed**erated **c**ognitive **d**iagnosis framework (FedCD) to jointly train CD models built upon a novel parameter decoupling-based personalization strategy, preserving privacy of data and achieving precise and fair diagnosis of students on each client. As an FL paradigm, FedCD trains a local CD model for the students in each client based on its local student learning data, and each client uploads its partial model parameters to the central server for parameter aggregation according to the devised innovative personalization strategy. The main idea of this strategy is to decouple model parameters into two parts: the first is used as locally personalized parameters, containing diagnostic function-related model parameters, to diagnose each client's students fairly; the second is the globally shared parameters across clients and the server, containing exercise embedding parameters, which are updated via fairness-aware aggregation, to alleviate inter-school unfairness. Experiments on three real-world datasets demonstrate the effectiveness of the proposed FedCD framework and the personalization strategy compared to five FL approaches under three CD models.

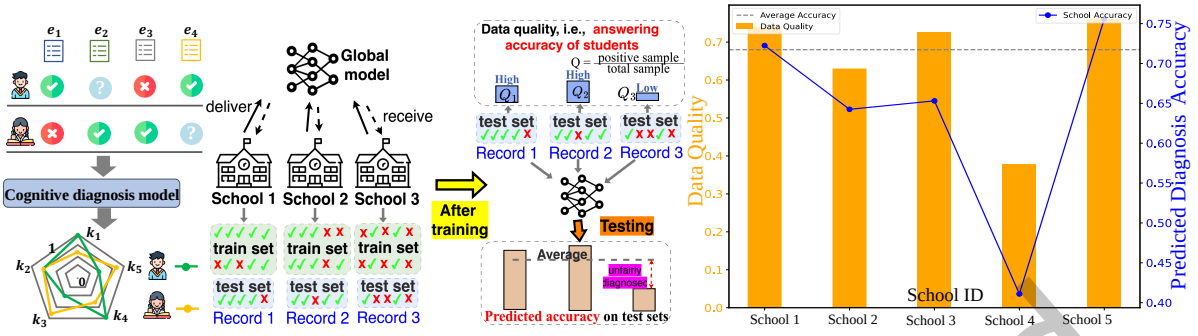
CCS Concepts: • **Information systems → Intelligent education systems**; • **Applied computing →** *E-learning*; • **Computing methodologies → Federated learning**.

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(a) The CD process (b) The client fairness issue for FL in education (c) Case study of the fairness issue

Fig. 1. (a) A toy example of CD. (b) The client fairness issue: different schools may exhibit heterogeneous response distributions, e.g., substantially different correct-response rates, and common FL approaches may consequently provide imbalanced diagnosis performance across school clients, such as School 3. (c) A case study by implementing FedAvg and NCD on five schools of students within the ASSISTment2009 dataset.

Additional Key Words and Phrases: Intelligent education, educational data mining, learner modeling, cognitive diagnosis, fairness-aware learning, federated learning

1 Introduction

In recent years, many online intelligent education platforms (OIDPs) have emerged with significant growth, such as JunYi [33], Khan [1], Coursera [5], MOOC [42], and PTA [16]. These platforms have attracted many users (i.e., students or learners) and have generated massive amounts of various learning data [39]. By collecting and analyzing these distributed learning data, OIDPs can assess the abilities of users [29] and then provide them effective learning recommendation services [46], including exercise recommendation [61], course recommendation [56], and learning path recommendation [62], to further enhance their abilities on poorly mastered knowledge concepts. The process of diagnosing students' mastery on specific knowledge concepts is called cognitive diagnosis (CD) [48], which is a fundamental upstream technique but plays a pivotal role in OIDPs because the accuracy of the diagnosis results profoundly influences the downstream recommendation quality [11]. For better understanding, Fig. 1(a) gives a toy example of CD: two students' knowledge mastery of five knowledge concepts is obtained by modeling their answer/response prediction.

To model these collected historical learning data of users, researchers have developed a series of cognitive diagnosis models (CDMs) based on various ideas [47], which can be categorized into three types. The first type focuses on the model interpretability and thus creates CDMs based on the theory of educational psychology [48], where IRT [9], MIRT [41], and DINA [45] are the representative approaches. Considering the powerful modeling ability of neural networks (NNs) [3], the second genre aims to incorporate NNs with the above CDMs to enhance diagnosis accuracy, where the representatives include NCD [48], KaNCD [49], KSCD [33] and RCD [11]. To save human labor on manually designing CDMs, the third leverages the idea of automated machine learning [19] like neural architecture search [44, 60, 73] to search CDMs, with EMO-NAS-CD [58, 59] and NAS-GCD [62] as the representative.

Although CDMs achieved considerable success, they rely on collecting all users' learning data together for centralized training [72]. The existing CDMs are effective in terms of model training but have been criticized for data privacy and security risks [70], because user data is often highly sensitive but centralized storage and

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processing raise privacy concerns and increase data leakage risks [31]. Therefore, many users are reluctant to share their data, limiting the practicality of the current paradigm of CDMs [54]. To achieve the data privacy-preserved cognitive diagnosis paradigm, an intuitive idea is to leverage the federated learning (FL) [21] for CD. FL enables users to keep their data on local clients without sharing it and trains models by only communicating model parameters between user clients and the central server. FL has made significant progress in many tasks [25], such as recommendation [57], medical image [2] and Internet of Things [36], but has not been explored in CD to protect user privacy.

To fill this gap, this paper seeks to integrate FL with CD to ensure the diagnosis accuracy of CDMs while preserving user privacy. Data heterogeneity is one well-known challenge in federated learning [40, 43, 50, 72]. When FL is applied to CD, this general challenge manifests as substantial differences in response distributions and data volumes across school clients. In this work, we use *response-distribution heterogeneity* to describe inter-school differences in the proportions of correct and incorrect responses, rather than interpreting a lower correct-response rate as intrinsically lower data quality. Specifically, differences in student populations, educational contexts, and learning behaviors can lead to different response distributions across school clients. We use the client-level correct-response rate as an observable summary of such response distributions and separately characterize data-volume heterogeneity in the experimental section. As shown in Fig. 1(b), students from different schools may differ in their educational contexts and learning characteristics, such as course schedules, available educational resources, and initial proficiency levels. These differences can result in heterogeneous local response distributions; for example, School 3 exhibits a substantially lower correct-response rate than the other schools. As a result, CDMs trained by common FL approaches may provide considerably different diagnosis performance across school clients. Consequently, clients whose local response distributions differ substantially from those of other clients may receive considerably poorer diagnosis performance under common FL approaches, leading to a client fairness issue, as illustrated by School 3 in Fig. 1(b). To validate the client fairness issue, an easy case study implementing NCD with FedAvg [27] is performed on five schools of students in ASSISTment2009 [10]. In Fig. 1(c), School 4 exhibits a much lower correct-response rate than the other schools, indicating a substantially different local response distribution. Meanwhile, the NCD model trained by FedAvg obtains considerably lower diagnostic accuracy on this client than on the others. This observation intuitively illustrates that heterogeneous client response distributions can be accompanied by substantial client-level performance disparity in federated CD. Although HPFL [55] and AHPFL [30] employ the FL technique for CD, they aim to solve the heterogeneous data within clients but not to address the client fairness issue, leading to limited scalability.

To address the above issues, we propose a fairness-aware federated cognitive diagnosis framework (FedCD) to preserve the privacy of student learning data and accurately diagnose students in each client. Specifically, our contributions are summarized below:

- To tackle the client fairness issue under heterogeneous client response distributions, a novel parameter decoupling-based personalization strategy is devised for FedCD to provide more balanced and accurate diagnosis across school clients. Following the fundamental procedures of common FL approaches, the proposed FedCD trains a local CDM for each client using local student learning data and aggregates model parameters in the central server to update parameters and assign them to clients, repeating multiple rounds.
- The proposed parameter decoupling-based personalization strategy divides the model parameters in each client into two parts: a locally personalized part and a globally shared part across all clients and the server. The former contains the diagnostic function-related model parameters to capture the local student characteristics. The latter contains exercise embedding parameters, updated by fairness-aware aggregation, which was designed to mitigate the school fairness issue.

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- Extensive experiments demonstrate the effectiveness of the proposed FedCD compared with five FL approaches under three CDMs on three real-world datasets, as well as the effectiveness of the proposed personalization strategy. Moreover, client fairness is systematically evaluated from group-level disparity, overall client-performance dispersion, and worst-client performance, showing that FedCD can reduce the performance disparity among heterogeneous clients while maintaining promising diagnostic utility.

2 Related Work

2.1 Cognitive Diagnosis

As one of many user modeling techniques [18], cognitive diagnosis aims to extract students' features from their large amounts of historical learning data, similar to recommender systems [13]. However, due to the nature of the CD task, the extracted features must be interpretable. Thus, existing CDMs represent each student's features through a single ability scalar or a latent vector that indicates their mastery level of each knowledge concept. As two pioneering approaches, IRT [9] and DINA [45] represent student mastery using a continuous scalar and a binary vector, respectively, where the answer prediction process is completed using a logistic function in IRT and an inner-product operation in DINA. Inspired by IRT and DINA, later-developed CDMs adopt the idea of using a continuous vector to represent students' knowledge proficiency or states, but these CDMs employ different techniques for diagnosis, including matrix factorization-based MF [23], multilayer perceptron-based NCD [48], KaNCD [49], KSCD [33], graph NN-based RCD [12], CDGK [52], SCD [51], Bayesian network-based HierCDF [24], inductive learning-based ICDM [32], and others [58, 62]. Despite the variety of techniques in existing CDMs, their effectiveness largely depends on the centralized training of models using all students' data, raising concerns about data privacy and potential data leakage [21]. Fortunately, a federated approach, HPFL [55] (AHPFL [30] is an extension of HPFL), uses a hierarchical local model to tackle the data heterogeneity problem. However, HPFL relies heavily on a fixed model, GUM, which limits its scalability and applicability to other CDMs. More importantly, HPFL does not address client fairness issues. Therefore, it is urgent to develop a scalable CD paradigm to ensure data privacy and fairness.

2.2 Federated Learning

As one of the most promising techniques, federated learning employs a central server to coordinate multiple clients in collaboratively training a model without directly accessing their local data [22]. To effectively aggregate model parameters, researchers have proposed various strategies, such as FedAvg [34], FedAtt [20], FedSGD [35], and LoAdaBoost [17]. Beyond model aggregation, data heterogeneity and distribution imbalance across clients have been widely recognized as important challenges in federated learning. In particular, Wang et al. [50] investigated class imbalance in federated learning, while Shuai et al. [43] further considered class imbalance under long-tailed federated data. More recently, Qi et al. [40] studied imbalanced client data from the perspective of cross-silo feature-space inconsistency and introduced feature-space alignment for heterogeneous clients. These studies demonstrate that distribution imbalance is a general challenge in federated learning and has been investigated from different perspectives, such as class distributions and feature representations.

Based on these fundamental FL approaches, federated learning has been widely extended to various domains and has achieved great success, including banking [65], healthcare [2], personal devices [64], and many others [25]. Among these, FL-based recommendation [67, 68], also known as federated recommendation (FedRec), is most closely related to cognitive diagnosis. The representative FedRec approaches include MetaMF [28], which uses a meta-network to generate private item embeddings and a rating prediction model; FedNCF [38], which applies the classical NCF [14] approach to the federated setting; FedPerGNN [53], which learns high-order user-item information through graph NNs; and PFedRec [67], which incorporates a parameter personalization mechanism.

Inspired by the success of FL in these domains, recent studies have also explored federated learning in intelligent education. For example, Wu et al. [54] investigated federated student modeling and discussed substantial

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differences among different schools or groups of students. Such observations indicate that the general data-heterogeneity problem in FL also exists in intelligent education, where different school clients may exhibit substantially different response distributions and data volumes. Nevertheless, the above general FL studies and the problem considered in this work focus on different aspects of heterogeneous federated data. Existing imbalance-oriented FL approaches mainly address issues such as class-distribution imbalance, long-tailed data, or feature-space inconsistency, with the primary goal of improving predictive performance under heterogeneous client distributions [40, 43, 50]. In federated CD, each school serves as a client and its local response data jointly involve students, exercises, and knowledge concepts. Consequently, inter-school response-distribution heterogeneity may further lead to substantial disparities in diagnosis performance among school clients. Although some fair FL approaches have been proposed [31, 66], they are not specifically designed for this CD-specific modeling structure and client-fairness objective. Therefore, the contribution of FedCD is not to identify data heterogeneity as a new problem in FL, but to address its manifestation as a client-fairness issue in federated cognitive diagnosis.

2.3 Difference between Federated Recommendation and Federated Cognitive Diagnosis

Federated recommendation is an emerging paradigm that enables collaborative filtering and personalized content delivery while preserving user privacy through decentralized learning. It typically models both users and items to infer users' preferences for unseen items, aiming to predict the likelihood of user-item interactions [4]. While both CD and recommendation systems utilize latent user representations, they differ substantially in data structure, objectives, and modeling paradigms. CD involves a multi-entity structure, i.e. students, exercises, knowledge concepts, and schools (as clients) [55], leading to intra-client data imbalance, a challenge not present in typical recommendation systems where each client represents a single user. CD aims to assess student mastery over predefined knowledge concepts, with the diagnosis results explicitly aligned with the Q-matrix and educational constraints such as monotonicity [71]. In contrast, recommendation primarily focuses on modeling user–item preferences and predicting user–item interactions. Although model interpretability has also been actively studied in recommender systems, e.g., through post-hoc explanation mining for latent factor models [37], its explanatory target is different from that of CD. Specifically, CD directly provides concept-level proficiency estimates for students, whereas recommendation explanations mainly aim to interpret user preferences or recommendation outcomes. Furthermore, CD functions as a core diagnostic task in educational systems, supporting downstream applications like personalized learning paths [26]. Existing federated recommendation methods often underperform in CD settings due to these fundamental differences [55], making it inappropriate to treat CD as a subset of recommendation.

3 Preliminaries

3.1 Cognitive Diagnosis

For the CD task in one online intelligent education platform, three sets of items are generally collected together for prediction modeling, including a set of M exercises $\mathcal{E} = \{e_1, e_2, \ldots, e_M\}$, a set of N students $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$, and a set of K knowledge concepts (concepts for short) $\mathcal{C} = \{c_1, c_2, \ldots, c_K\}$. In addition, a prior exercise-concept relation matrix (called Q-matrix) is also needed, denoted as $Q = (Q_{jk} \in \{0, 1\})^{M \times K}$, where $Q_{jk} = 1$ refers to the concept c_k included in the exercise e_j and $Q_{jk} = 0$ otherwise.

In traditional centralized training of CDMs, all students' learning data (known as exercising response logs) need to be collected together as a set of triplets $\mathcal{R} = \{(s_i, e_j, r_{ij}) | s_i \in \mathcal{S}, e_j \in \mathcal{E}, r_{ij} \in \{0, 1\}\}$. Here r_{ij} refers to the response/answer of student s_i on exercise e_j. $r_{ij} = 1$ means the answer is correct and $r_{ij} = 0$ otherwise. As a result, **given** students' response logs $\mathcal{R}$ and the Q-matrix, the **goal** of CD is to reveal students' proficiency on concepts

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by training a CDM $\mathcal{M}(\cdot|\theta)$ to model the students' answer prediction as follows:

$$\min_{\theta\in\Theta} \sum_{(s_i,e_j,r_{ij})\in\mathcal{R}} Loss\left(r_{ij}, \mathcal{M}(\mathrm{x}_i^S, \mathrm{x}_j^E|\theta)\right)$$
$$\mathcal{M}(\mathrm{x}_i^S, \mathrm{x}_j^E|\theta) = \begin{cases} \mathrm{h}_S = Emb_1(\mathrm{x}_i^S|\theta^S),\ \mathrm{h}_E = Emb_2(\mathrm{x}_j^E|\theta^E) \\ \mathrm{h}_C = \mathrm{x}_j^E \times Q = (Q_{j1}, Q_{j2}, \cdots, Q_{jK}) \\ \hat{r}_{ij} = \mathcal{J}(\mathrm{h}_S, \mathrm{h}_E, \mathrm{h}_C|\theta^I) \end{cases}. \tag{1}$$

Here $\mathrm{x}_i^S \in \{0,1\}^{1\times N}$ and $\mathrm{x}_j^E \in \{0,1\}^{1\times M}$ are one-hot vectors for student s_i and exercise e_j, respectively. The student-related vector $\mathrm{h}_S \in \mathbb{R}^{1\times D}$ and exercise-related vector $\mathrm{h}_E \in \mathbb{R}^{1\times D}$ are obtained by two embedding modules $Emb_1(\cdot|\theta^S)$ and $Emb_2(\cdot|\theta^E)$, parameterized by θ^S and θ^E. $\mathcal{J}(\cdot|\theta^I)$ denotes the diagnostic function to obtain the predicted answer $\hat{r}_{ij}$ of student s_i and exercise e_j, which is parameterized by θ^I. As a result, the model parameters θ of $\mathcal{M}(\cdot|\theta)$ contains three components, i.e., $\theta = \{\theta^S, \theta^E, \theta^I\}$.

After optimizing θ to minimize the loss in the first item in Eq.(1), h_S can be used to represent the student's knowledge proficiency, which is the diagnosis result of student s_i.

3.2 Federated Learning

Different from previous CDMs, FL learns a global model parameterized by $\bar{\theta}$ to serve all clients of users without directly accessing their data. As a result, the target of a common FL approach is to minimize the loss of all T clients as follows:

$$\min_{\bar{\theta}\in\Theta} \sum_{t=1}^{T} \alpha_t \mathcal{L}\left(D_t|\bar{\theta}\right), \tag{2}$$

where $\mathcal{L}\left(D_t|\bar{\theta}\right)$ denotes the supervised loss on the t-th client with its dataset D_t. α_t is the weight for the t-th client's loss. Taking FedAvg [34] as an example, its α_t is defined as the fraction of t-th client's data size, i.e., $\alpha_t = |D_t| / \sum_t^T |D_t|$. After the global model is trained, it can be utilized to predict tasks for all users' clients.

3.3 Federated Cognitive Diagnosis Definition

In online intelligent education, suppose there are T schools/groups of students $\mathcal{S} = \left\{\mathcal{S}_{T_t} = \{s_t^1, s_t^2, \cdots, s_t^{N_t}\} | 1 \le t \le T\right\}$ and the t-th school has N_t students as $\mathcal{S}_{T_t} = \{s_t^1, s_t^2, \cdots, s_t^{N_t}\}$ is regarded as the t-th client. Therefore, the equation $\sum_t^T N_t = N$ holds. Then, the learning data contained in t-th client is denoted as $\mathcal{R}_{T_t} = \left\{(s_i, e_j, r_{ij}) | s_i \in \mathcal{S}_{T_t}, e_j \in \mathcal{E}, r_{ij} \in \{0,1\}\right\}$. **Given** the above information, the goal of federated CD is to reveal students' knowledge states by minimizing the diagnosis loss of all T clients:

$$\min_{\bar{\theta}\in\Theta} \left(\sum_{t=1}^{T} \alpha_t \times \sum_{(s_i,e_j,r_{ij})\in\mathcal{R}_{T_t}} \mathrm{Loss}\left(r_{ij}, \mathcal{M}(\mathrm{x}_i^S, \mathrm{x}_j^E \mid \bar{\theta})\right) \right), \tag{3}$$

where $\mathcal{M}(\mathrm{x}_i^S, \mathrm{x}_j^E|\bar{\theta})$ is same as that in Eq.(1). Here $\bar{\theta} = \{\bar{\theta}^S, \bar{\theta}^E, \bar{\theta}^I\}$ are all the same for all clients in common FL approaches.

Besides overall diagnostic utility, this work particularly considers client fairness in federated CD. Specifically, client fairness refers to reducing the diagnosis performance disparity among different school clients, particularly preventing clients whose local response distributions differ substantially from those of other clients from receiving considerably worse diagnosis performance, while maintaining promising overall diagnostic utility. Accordingly,

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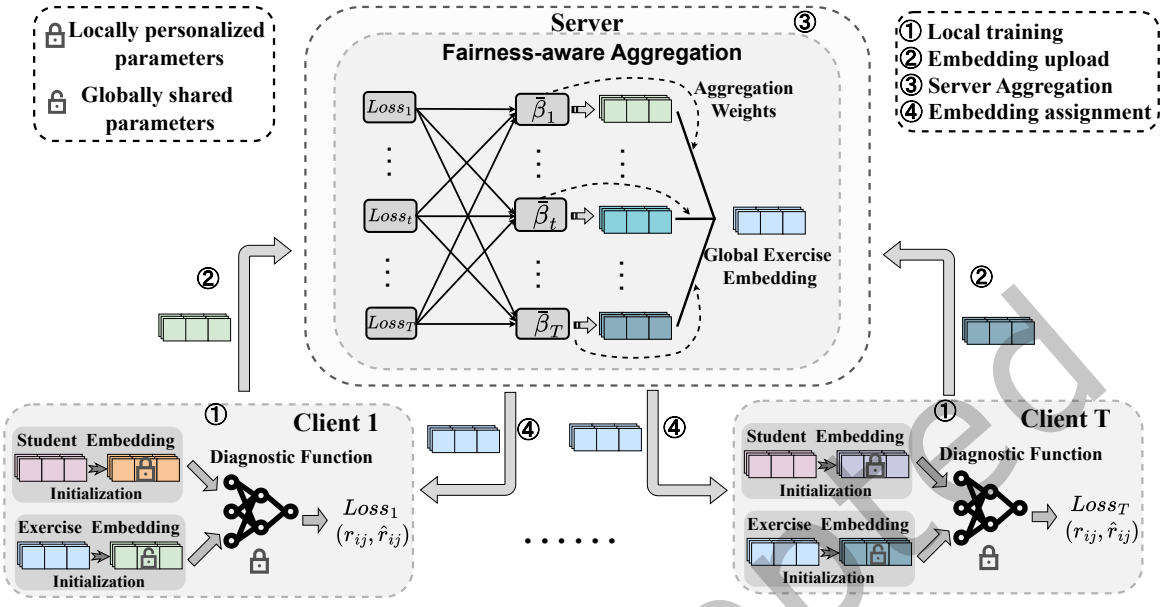


Fig. 2. Overview of FedCD. Each communication round consists of four steps: ① Each school/client trains its own CDM, initializing its local exercise embedding with the globally shared embeddings. These are then substituted into the locally preserved student embedding and diagnostic function to compute the predicted loss $Loss_t(y, \hat{y})$. ② Each client uploads its exercise embedding to the server. ③ The server determines the aggregation weights based on the chosen federated framework and aggregates the local exercise embeddings from each client to obtain the global exercise embedding. ④ Finally, the server distributes the global exercise embedding back to the clients.

FedCD aims not only to achieve accurate diagnosis but also to obtain more balanced diagnosis performance across clients.

4 The Proposed FedCD Framework

4.1 Overview

In this section, we present the fairness-aware federated cognitive diagnosis framework. As shown in Fig. 2, the client fairness issue is addressed by implementing a parameter decoupling-based personalization strategy, which separates the t-th client's model parameters $\theta_t = \{\theta_t^S, \theta_t^E, \theta_t^I\}$ into two parts. The first part is referred to as the **globally shared parameters**, which include the parameters of exercise embedding modules, Emb_2, i.e., $\{\theta_t^E\}$. The second part is known as the **locally personalized parameters**, which pertain to the parameters of the diagnostic function $\mathcal{J}(\cdot)$ and student embedding modules, i.e., $\{\theta_t^S, \theta_t^I\}$. In summary, the central server only updates the parameters $\bar{\theta}^E$ as $\bar{\theta}$ from all clients, i.e., $\bar{\theta} = \{\bar{\theta}^E\}$, while each client retains its personalized diagnostic function parameters $\bar{\theta}^I$ and student embedding parameters $\bar{\theta}^S$, i.e., $\{\theta_t^S, \theta_t^I\}$. The intuition behind this decoupling is that student characteristics and diagnostic patterns are highly client-specific, whereas exercises and their associated knowledge concepts are defined in a common space across clients and can provide transferable cross-client knowledge. Therefore, FedCD keeps student-side diagnostic information locally personalized while sharing exercise-related representations across clients. With this strategy, the procedures for each training round of the

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FedCD framework are as follows: First, the clients train their local models based on their own student learning data. Second, each client keeps the parameters θ_t^S and θ_t^I local for the next round of training and uploads the parameters θ_t^E to the server. Next, the server updates the exercise embedding parameters $\tilde{\theta}^E$ through a simple fairness-aware weighted aggregation mechanism. Finally, the exercise embedding parameters are assigned to all clients for local model training.

In the following part, we provide a detailed introduction to the proposed FedCD. First, we explain the parameter decoupling-based personalization strategy by outlining the design of the optimization objective, client, and server, respectively. Then, we describe the learning process and summarize the overall optimization workflow in the form of an algorithm. Finally, we examine the compatibility of FedCD with an existing local privacy-preserving technique by integrating a standard noise-perturbation mechanism as an optional extension.

4.2 Parameter Decoupling-based Personalization Strategy

The parameter decoupling in FedCD is motivated by the different cross-client characteristics of student-side and exercise-side information. For the student side, students in different schools are naturally different and may exhibit substantial differences in ability, learning behavior, educational context, and response distributions. Therefore, globally sharing the student embeddings and diagnostic function parameters may force heterogeneous client-specific diagnostic patterns into a unified model. FedCD thus keeps $\{\theta_t^S, \theta_t^I\}$ locally personalized so that each client can adapt its diagnosis to its own students.

In contrast, exercises and their associated knowledge concepts are defined in a common exercise-concept space across clients in our federated CD setting. Although students from different schools may respond differently to the same or related exercises, their response records provide complementary information for learning exercise-related representations. For clients with limited local response observations or response distributions that differ substantially from those of other clients, estimating exercise embeddings only from local responses may result in less reliable representations. Therefore, FedCD globally shares the exercise embedding parameters θ_t^E, allowing each client to benefit from exercise-related knowledge learned across different schools. Moreover, the globally shared exercise embeddings are updated through the fairness-aware aggregation in Eq. (8), where clients with larger diagnosis losses are assigned larger aggregation weights. This prevents the shared representations from being dominated by well-performing clients and gives poorly performing clients greater influence during global knowledge aggregation.

Hence, the two parts play complementary roles: locally personalized student embeddings and diagnostic functions preserve client-specific diagnostic characteristics, while globally shared exercise embeddings provide cross-client collaborative knowledge. Together with fairness-aware aggregation, this parameter decoupling is designed to improve diagnosis performance while alleviating performance disparity among clients.

4.2.1 Optimization Objective Design. Due to the locally personalized parameters of the model in each client, the overall optimization objective of FedCD will be turned to the following:

$$\begin{aligned} &\min_{\{\tilde{\theta}^E \in \Theta^E\}} \sum_{t=1}^{T} \alpha_t \times \\ &\sum_{(s_i, e_j, r_{ij}) \in \mathcal{R}_{T_t}} Loss\left(r_{ij}, \mathcal{M}(\mathrm{x}_i^S, \mathrm{x}_j^E | \{\theta_t^S, \theta_t^E, \theta_t^I\})\right), \\ s.t.,\ &\theta_t^S, \theta_t^E, \theta_t^I = \underset{\{\hat{\theta}_t^S \in \Theta^S, \hat{\theta}_t^E \in \Theta^E, \hat{\theta}_t^I \in \Theta^I\}}{\arg\min} \\ &\sum_{(s_i, e_j, r_{ij}) \in \mathcal{R}_{T_t}} Loss\left(r_{ij}, \mathcal{M}(\mathrm{x}_i^S, \mathrm{x}_j^E | \{\hat{\theta}_t^S, \hat{\theta}_t^E, \hat{\theta}_t^I\})\right), \end{aligned} \tag{4}$$

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where Θ^S, Θ^E, and Θ^I denote the parameter search space. The above equation is a bi-level optimization problem: the upper-level problem solved in the server aims to find optimal $\bar{\theta}^E$ as the initial parameters for the lower-level problem in each client, i.e., $\hat{\theta}_t^E$ takes $\bar{\theta}^E$ as initial values; while the lower-level problem aims to find optimal parameters $\{\theta_t^S, \theta_t^E, \theta_t^I\}$ by minimizing the loss values on the data $\mathcal{R}_{T_t}$. It is important to note that solving the upper-level problem requires multiple rounds. In each round, $\hat{\theta}_t^S$ and $\hat{\theta}_t^I$ in the lower-level problem use the θ_t^S and θ_t^I in the last round as the initial value, while in the first round, $\hat{\theta}_t^S$ and $\hat{\theta}_t^I$ are initialized randomly.

The loss weight of t-th client α_t is set to 1 to consider client fairness, ignoring the data size influence. The forward-pass process of $\mathcal{M}(\mathrm{x}_i^S, \mathrm{x}_j^E | \{\hat{\theta}_t^S, \hat{\theta}_t^E, \hat{\theta}_t^I\})$ is same as that in Eq.(1), which outputs the predicted answer of student s_i got on exercise e_j, i.e., $\hat{r}_{ij}$. The binary cross-entropy loss [63] is used to measure the loss between $\hat{r}_{ij}$ and r_{ij} as follows:

$$Loss(r_{ij}, \hat{r}_{ij}) = -\left(r_{ij} log(\hat{r}_{ij}) + (1 - r_{ij}) log(1 - \hat{r}_{ij})\right). \tag{5}$$

4.2.2 Client Design. The FedCD needs to train T models for T schools of students independently. These T models have the same network architecture but different weight parameters, where the model parameters of t-th client are $\theta_t = \{\theta_t^S, \theta_t^E, \theta_t^I\}$.

For most existing CDMs, θ_t^S and θ_t^E are the same according to Eq.(1), and the student embedding module $Emb_1(\cdot)$ and the exercise embedding module $Emb_2(\cdot)$ in the t-th client can be denoted as

$$\begin{aligned} \mathrm{h}_S &= Emb_1(\mathrm{x}_i^S | \theta_t^S) = x_i^S \times \theta_t^S,\ \theta_t^S \in \mathbb{R}^{N \times D} \\ \mathrm{h}_E &= Emb_2(\mathrm{x}_j^E | \theta_t^E) = x_j^E \times \theta_t^E,\ \theta_t^E \in \mathbb{R}^{M \times D} \end{aligned}, \tag{6}$$

where D is the embedding size, usually equal to K. θ_t^S and θ_t^E are two learnable parameter matrices to store all students' and all exercises' embedding parameters.

Generally, different CDMs have different architectures of $\mathcal{I}(\cdot)$ and thus θ_t^I is various. Here we take NCD as an example, whose forward-pass process is as follows:

$$\mathcal{I}(\mathrm{h}_S, \mathrm{h}_E, \mathrm{h}_C | \theta_t^I) = \begin{cases} \mathrm{f}_S = \sigma(\mathrm{h}_S),\ \mathrm{f}_{diff} = \sigma(\mathrm{h}_E) \\ f_{disc} = \sigma(\mathrm{h}_E \times W_{disc}), W_{disc} \in \mathbb{R}^{D \times 1} \\ \mathrm{y} = \mathrm{h}_C \odot (\mathrm{f}_S - \mathrm{f}_{diff}) \times f_{disc} \\ \hat{r}_{ij} = \sigma(\sigma(\sigma(\mathrm{y} \times W_{fc1}) \times W_{fc2}) \times W_{fc3}) \end{cases}, \tag{7}$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, $\mathrm{f}_{diff} \in \mathbb{R}^{1 \times D}$ and $\mathrm{f}_{disc} \in \mathbb{R}^1$ represent the difficulty and the discrimination of the exercise. The last item denotes the employed three-layer fully connected layers, whose three parameters are $W_{fc1} \in \mathbb{R}^{D \times 4D}$, $W_{fc2} \in \mathbb{R}^{4D \times 2D}$, and $W_{fc3} \in \mathbb{R}^{2D \times 1}$. As a result, the diagnostic function-related model parameters for t-th client can be represented as $\theta_t^I = \{W_{disc}, W_{fc1}, W_{fc2}, W_{fc3}\}$.

As shown in Eq.(4), equipped with the server-assigned $\bar{\theta}^E$ and locally-saved $\{\theta_t^S, \theta_t^I\}$ as initial $\hat{\theta}_t^E$ and $\{\hat{\theta}_t^S, \hat{\theta}_t^I\}$, each client's model needs to be trained on data $\mathcal{R}_{T_t}$ for Num epochs to get its promising model parameters $\theta_t = \{\theta_t^S, \theta_t^E, \theta_t^I\}$. After that, θ_t^E is uploaded to the server to generate new parameters for the next round, and $\{\theta_t^S, \theta_t^I\}$ is kept local to be used directly as the initial of the next round.

4.2.3 Server Design. After all clients' models are locally trained, the central server is responsible for aggregating their uploaded parameter sets, i.e., exercise embedding parameter set $\{\theta_1^E, \cdots, \theta_T^E\}$.

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Algorithm 1 Algorithm of FedCD

1: Randomly initialize $\bar{\theta}^E$ and each client's local $\{\theta_t^S, \theta_t^I\}_{t=1}^T$;
2: **for** each communication round $p = 1, 2, \dots, P$ **do**
3: **for** each client $t = 1, \dots, T$ **in parallel do**
4: Set $\hat{\theta}_t^E \leftarrow \bar{\theta}^E, \hat{\theta}_t^S \leftarrow \theta_t^S, \hat{\theta}_t^I \leftarrow \theta_t^I$;
5: Train $(\theta_t^S, \theta_t^E, \theta_t^I)$ for Num epochs by minimizing Eq.(5);
6: Upload θ_t^E and local loss $\mathcal{L}_t$; keep θ_t^S, θ_t^I locally;
7: **end for**
8: Server updates $\bar{\theta}^E$ by Eq.(8);
9: Server broadcasts $\bar{\theta}^E$ to all clients;
10: **end for**

To obtain new exercise embedding parameters $\bar{\theta}^E \in \mathbb{R}^{M \times D}$, the following fairness-aware aggregation mechanism is adopted:

$$\begin{aligned} \bar{\theta}^E &= \sum_{t=1}^{T} \left(\bar{\beta}_t \times \theta_t^E\right), \text{ where } \bar{\beta}_t = \frac{exp(\gamma \cdot \mathcal{L}_t)}{\sum_{j=1}^{T} exp(\gamma \cdot \mathcal{L}_j)}, \\ \mathcal{L}_t &= \sum_{(s_i, e_j, r_{ij}) \in \mathcal{R}_{T_t}} Loss\left(r_{ij}, \mathcal{M}(\mathrm{x}_i^S, \mathrm{x}_j^E | \{\theta_t^S, \theta_t^E, \theta_t^I\})\right), \end{aligned} \tag{8}$$

where $\bar{\beta}_t \in \mathbb{R}^1$ is a normalized weight of t-th client's parameters to obtain $\bar{\theta}^E$. Here γ is a parameter adjusting the importance of the client loss $\mathcal{L}_t$. In short, its main idea is to weighted-sum all clients' parameters according to the influence of their loss values, where a larger loss value represents a larger weight for updating $\bar{\theta}^E$.

4.3 Algorithm Overall Procedure

The overall procedure of the proposed FedCD framework is summarized in Algorithm 1. First, the server randomly initializes the global exercise embedding $\bar{\theta}^E$, and each client randomly initializes its local student embedding θ_t^S and diagnostic function parameters θ_t^I (Line 1). In each communication round, each client sets $\hat{\theta}_t^E \leftarrow \bar{\theta}^E, \hat{\theta}_t^S \leftarrow \theta_t^S$, and $\hat{\theta}_t^I \leftarrow \theta_t^I$ (Line 4), and then trains its local model for Num epochs by minimizing the loss in Eq. (5) (Line 5). After local training, each client uploads its exercise embedding θ_t^E and local loss $\mathcal{L}_t$ to the server, while keeping θ_t^S and θ_t^I locally (Line 6). The server then updates $\bar{\theta}^E$ by applying the aggregation mechanism in Eq. (8) to the collected $\{\theta_1^E, \dots, \theta_T^E\}$ and $\{\mathcal{L}_1, \dots, \mathcal{L}_T\}$ (Line 8), and broadcasts $\bar{\theta}^E$ to all clients (Line 9). The above client training and server aggregation steps are repeated until convergence.

4.4 Integration with Local Differential Privacy

The proposed FedCD itself does not introduce a new differential privacy mechanism. Rather, to examine whether FedCD can be readily combined with additional privacy-preserving techniques, we adopt the commonly used Laplace-noise perturbation mechanism from differential privacy [8, 69] as an optional extension. Similar noise-perturbation techniques have also been employed in federated learning and federated recommendation to protect transmitted client information [31]. Specifically, before the t-th client uploads its locally trained exercise embedding parameters θ_t^E to the server, element-wise zero-mean Laplace noise is added as

$$\tilde{\theta}_t^E = \theta_t^E + \mathrm{Z}_t, \qquad [\mathrm{Z}_t]_{m,d} \sim \text{Laplace}(0, \delta), \tag{9}$$

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where Z_t has the same dimension as θ_t^E, and δ controls the magnitude of the perturbation. The perturbed parameters $\tilde{\theta}_t^E$ are then uploaded and used for server-side aggregation instead of the original θ_t^E. A larger δ introduces stronger perturbation to the transmitted parameters. This mechanism is employed only as an existing privacy-preserving extension of FedCD rather than as a methodological contribution of this work. Its influence on the diagnostic utility of FedCD is empirically evaluated in experiments.

5 EXPERIMENTS

This section aims to answer the following research questions:

- **RQ1:** How about the effectiveness of the proposed FedCD compared to classical FL approaches under different CDMs?
- **RQ2:** How about the effectiveness of the devised parameter decoupling-based personalization strategy?
- **RQ3:** Can FedCD reduce diagnosis performance disparity among clients and improve client fairness?
- **RQ4:** How about the sensitivity of the proposed FedCD to key hyperparameter changes?
- **RQ5:** How about the robustness of the proposed FedCD when integrating local differential privacy?
- **RQ6:** Are there other interesting discussions about FedCD?

5.1 Experimental Setting

5.1.1 Datasets. To implement the FL paradigm, datasets must contain information on students' schools or groupings. Our experiments were conducted on three real-world datasets: ASSIST2009, ASSIST2012, and SLP-Math. Both ASSISTs (short for Assistments) and SLP-Math are public datasets that record students' learning logs from an online education system. During preprocessing, we first removed records with null values and duplicates. We then filtered out students with fewer than 5 records and schools with fewer than 1,000 records. More detailed descriptions and statistics are provided below and in Table 1.

- The ASSIST2009 dataset captures students' learning logs from the 2009-2010 "Non-skill Builder" period in an online education system. After preprocessing, the dataset comprises 324,659 learning logs from 3,180 students across 21 schools. These logs involve 17,560 exercises and cover 123 distinct concepts.
- The ASSIST2012 dataset contains students' learning data from the 2012-2013 School Data with Affect. Following preprocessing, this dataset includes 22,470 students and 52,447 exercises, spanning 263 distinct concepts.
- For SLP-Math, we selected the students' math scores, combining the Learner's Personal Info and School Info to form a dataset. The SLP-Math dataset, which includes 10 schools, contains 47,816 records from 1,035 students, with 544 exercises covering 31 distinct concepts.

To quantitatively characterize client heterogeneity without interpreting it as intrinsic data quality, we consider two complementary aspects: response-distribution heterogeneity and data-quantity heterogeneity. For the t-th client, its correct-response rate is calculated as

$$CR_t = \frac{1}{|\mathcal{R}_{T_t}|} \sum_{(s_i, e_j, r_{ij}) \in \mathcal{R}_{T_t}} r_{ij}. \tag{10}$$

Since r_{ij} is binary, CR_t provides an observable summary of the proportion of correct and incorrect responses within the t-th client. We emphasize that CR_t is used to characterize the local response distribution rather than to define the intrinsic quality of a client's data or the ability of its students.

We use the standard deviation of client correct-response rates,

$$H_{\text{resp}} = STD(CR_1, CR_2, \ldots, CR_T), \tag{11}$$

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Table 1. Statistics of ASSIST2009, ASSIST2012 and SLP-Math datasets

Statistics	ASSIST2009	ASSIST2012	SLP-Math
#of schools	21	141	10
#of records	324,659	2,570,541	47,816
#of students	3,180	22,470	1,035
#of exercises	17,560	52,447	544
#of knowledge concepts	123	263	31
Mean client CR	0.6109	0.6772	0.5394
H_{resp}	0.1198	0.0857	0.1112
Client CR range	0.2939–0.7661	0.3485–0.9805	0.3294–0.6978
H_{size}	1.9053	1.8915	0.5209

to characterize response-distribution heterogeneity across clients, where a larger H_{resp} indicates greater variation in client-level correct/incorrect response distributions. We additionally report the minimum–maximum range of CR_t to reflect extreme inter-client differences. For data-quantity heterogeneity, we calculate

$$H_{\text{size}} = \frac{STD(|\mathcal{R}_{T_1}|, \dots, |\mathcal{R}_{T_T}|)}{Mean(|\mathcal{R}_{T_1}|, \dots, |\mathcal{R}_{T_T}|)}. \tag{12}$$

The resulting statistics are reported in Table 1.

As shown in Table 1, all three datasets exhibit clear client-level response-distribution heterogeneity. The H_{resp} values are 0.1198, 0.0857, and 0.1112 on ASSIST2009, ASSIST2012, and SLP-Math, while their client correct-response rates span 0.2939–0.7661, 0.3485–0.9805, and 0.3294–0.6978. The datasets also exhibit different data-quantity heterogeneity, with H_{size} values of 1.9053, 1.8915, and 0.5209, respectively. These statistics show that the three datasets contain clear yet distinct patterns of naturally occurring client heterogeneity in both response distributions and data volumes.

These measures summarize differences in client-level correct-response rates and data volumes, but do not fully characterize distributional heterogeneity in federated CD. In particular, clients with similar correct-response rates and data sizes may still differ in their exercise distributions and knowledge-concept coverage or frequencies. Such differences are not directly captured by H_{resp} and H_{size}.

5.1.2 Baseline Approaches. Three CDMs and five FL approaches were employed to demonstrate the effectiveness of the proposed FedCD framework. Specifically, three representative CDMs were utilized as the base models under the FL paradigm to showcase the scalability of FedCD, including one classical model, DINA [45], and two state-of-the-art (SOTA) models, NCD [48] and KaNCD [49]. Two classical FL approaches, FedAvg [34] and FedAtt [20], one SOTA FL approach, LoAdaBoost [17], one personalization FL recommendation approach, PFedRec [67], and one fairness-aware FL recommendation approach, F2MF [31], were compared to highlight the superiority of the proposed FedCD. Additionally, the FL approach for CD, named HPFL [54], was also compared to demonstrate the effectiveness of FedCD in addressing the fairness issue. More details about the aforementioned CDMs and FL approaches can be found as follows.

- **DINA** models students' mastery as a multidimensional knowledge concept vector and considers the factors of students' guessing and slipping.

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Table 2. Performance of FedCD and all baseline approaches on ASSIST2009. **Bold:** the best, Underline: the runner-up. Results are reported as mean ± standard deviation over 10 independent runs. W/T/L denotes the Win/Tie/Loss results of the Wilcoxon rank-sum test at the 0.05 significance level

Base	Train/Test ratio	60%/40%			70%/30%			80%/20%		
Model	Models	ACC↑	RMSE↓	AUC↑	ACC↑	RMSE↓	AUC↑	ACC↑	RMSE↓	AUC↑
DINA	Base	$0.6557_{\pm 0.0006}$	$0.4768_{\pm 0.0004}$	$0.6899_{\pm 0.0014}$	$0.6591_{\pm 0.0005}$	$0.4755_{\pm 0.0002}$	$0.6937_{\pm 0.0013}$	$0.6602_{\pm 0.0011}$	$0.4749_{\pm 0.0004}$	$0.7048_{\pm 0.0012}$
	FedAvg	$0.6508_{\pm 0.0008}$	$0.4737_{\pm 0.0004}$	$0.6628_{\pm 0.0010}$	$0.6542_{\pm 0.0010}$	$0.4715_{\pm 0.0003}$	$0.6690_{\pm 0.0009}$	$0.6553_{\pm 0.0004}$	$0.4729_{\pm 0.0003}$	$0.6679_{\pm 0.0009}$
	FedAtt	$0.6526_{\pm 0.0005}$	$0.4763_{\pm 0.0003}$	$0.6506_{\pm 0.0058}$	$0.6545_{\pm 0.0009}$	$0.4731_{\pm 0.0003}$	$0.6653_{\pm 0.0011}$	$0.6568_{\pm 0.0003}$	$0.4738_{\pm 0.0002}$	$0.6671_{\pm 0.0006}$
	LoAdaBoost	$0.6359_{\pm 0.0008}$	$0.4795_{\pm 0.0003}$	$0.6642_{\pm 0.0019}$	$0.6360_{\pm 0.0028}$	$0.4795_{\pm 0.0010}$	$0.6651_{\pm 0.0033}$	$0.6345_{\pm 0.0029}$	$0.4802_{\pm 0.0009}$	$0.6680_{\pm 0.0027}$
	F2MF	$0.6518_{\pm 0.0008}$	$0.4759_{\pm 0.0002}$	$0.6499_{\pm 0.0011}$	$0.6545_{\pm 0.0007}$	$0.4715_{\pm 0.0001}$	$0.6680_{\pm 0.0005}$	$0.6549_{\pm 0.0008}$	$0.4729_{\pm 0.0002}$	$0.6680_{\pm 0.0013}$
	PFedRec	$0.6576_{\pm 0.0011}$	$0.4729_{\pm 0.0003}$	$0.6603_{\pm 0.0007}$	$0.6570_{\pm 0.0009}$	$0.4729_{\pm 0.0003}$	$0.6606_{\pm 0.0017}$	$0.6553_{\pm 0.0009}$	$0.4735_{\pm 0.0004}$	$0.6613_{\pm 0.0031}$
	FedCD	$0.6768_{\pm 0.0004}$	$0.4663_{\pm 0.0003}$	$0.7138_{\pm 0.0007}$	$0.6781_{\pm 0.0009}$	$0.4665_{\pm 0.0003}$	$0.7196_{\pm 0.0006}$	$0.6830_{\pm 0.0013}$	$0.4636_{\pm 0.0005}$	$0.7226_{\pm 0.0005}$
	FedCD(FedAvg)	$0.6728_{\pm 0.0011}$	$0.4673_{\pm 0.0004}$	$0.7096_{\pm 0.0008}$	$0.6734_{\pm 0.0011}$	$0.4676_{\pm 0.0003}$	$0.7148_{\pm 0.0014}$	$0.6813_{\pm 0.0009}$	$0.4643_{\pm 0.0003}$	$0.7201_{\pm 0.0009}$
	FedCD(FedAtt)	$0.6705_{\pm 0.0007}$	$0.4682_{\pm 0.0003}$	$0.7076_{\pm 0.0004}$	$0.6722_{\pm 0.0015}$	$0.4680_{\pm 0.0006}$	$0.7140_{\pm 0.0011}$	$0.6807_{\pm 0.0009}$	$0.4645_{\pm 0.0004}$	$0.7191_{\pm 0.0001}$
NCD	Base	$0.6797_{\pm 0.0017}$	$0.4494_{\pm 0.0002}$	$0.7086_{\pm 0.0005}$	$0.6868_{\pm 0.0025}$	$0.4488_{\pm 0.0016}$	$0.7096_{\pm 0.0048}$	$0.6929_{\pm 0.0022}$	$0.4478_{\pm 0.0009}$	$0.7181_{\pm 0.0010}$
	FedAvg	$0.6785_{\pm 0.0005}$	$0.4618_{\pm 0.0013}$	$0.6668_{\pm 0.0008}$	$0.6804_{\pm 0.0004}$	$0.4602_{\pm 0.0007}$	$0.6721_{\pm 0.0007}$	$0.6816_{\pm 0.0007}$	$0.4647_{\pm 0.0105}$	$0.6794_{\pm 0.0028}$
	FedAtt	$0.6744_{\pm 0.0014}$	$0.4699_{\pm 0.0011}$	$0.6624_{\pm 0.0012}$	$0.6752_{\pm 0.0017}$	$0.4691_{\pm 0.0005}$	$0.6660_{\pm 0.0010}$	$0.6755_{\pm 0.0024}$	$0.4706_{\pm 0.0063}$	$0.6722_{\pm 0.0019}$
	LoAdaBoost	$0.6935_{\pm 0.0007}$	$0.4511_{\pm 0.0003}$	$0.6982_{\pm 0.0004}$	$0.6938_{\pm 0.0019}$	$0.4505_{\pm 0.0007}$	$0.6998_{\pm 0.0006}$	$0.6978_{\pm 0.0005}$	$0.4502_{\pm 0.0003}$	$0.7045_{\pm 0.0006}$
	F2MF	$0.6781_{\pm 0.0003}$	$0.4619_{\pm 0.0015}$	$0.6668_{\pm 0.0012}$	$0.6806_{\pm 0.0007}$	$0.4595_{\pm 0.0007}$	$0.6714_{\pm 0.0011}$	$0.6817_{\pm 0.0004}$	$0.4597_{\pm 0.0003}$	$0.6778_{\pm 0.0004}$
	PFedRec	$0.6831_{\pm 0.0005}$	$0.4614_{\pm 0.0066}$	$0.6996_{\pm 0.0075}$	$0.6927_{\pm 0.0006}$	$0.4608_{\pm 0.0013}$	$0.7080_{\pm 0.0007}$	$0.6986_{\pm 0.0004}$	$0.4604_{\pm 0.0003}$	$0.7170_{\pm 0.0002}$
	FedCD	$0.7017_{\pm 0.0047}$	$0.4483_{\pm 0.0047}$	$0.7320_{\pm 0.0011}$	$0.7114_{\pm 0.0019}$	$0.4487_{\pm 0.0097}$	$0.7389_{\pm 0.0031}$	$0.7201_{\pm 0.0013}$	$0.4539_{\pm 0.0043}$	$0.7413_{\pm 0.0058}$
	FedCD(FedAvg)	$0.7013_{\pm 0.0016}$	$0.4502_{\pm 0.0025}$	$0.7257_{\pm 0.0016}$	$0.7054_{\pm 0.0026}$	$0.4452_{\pm 0.0008}$	$0.7325_{\pm 0.0011}$	$0.7097_{\pm 0.0018}$	$0.4423_{\pm 0.0013}$	$0.7430_{\pm 0.0032}$
	FedCD(FedAtt)	$0.7012_{\pm 0.0020}$	$0.4515_{\pm 0.0025}$	$0.7245_{\pm 0.0021}$	$0.7020_{\pm 0.0018}$	$0.4473_{\pm 0.0016}$	$0.7319_{\pm 0.0026}$	$0.7118_{\pm 0.0022}$	$0.4435_{\pm 0.0033}$	$0.7426_{\pm 0.0020}$
KaNCD	Base	$0.6975_{\pm 0.0123}$	$0.4621_{\pm 0.0130}$	$0.7376_{\pm 0.0034}$	$0.7024_{\pm 0.0042}$	$0.4605_{\pm 0.0109}$	$0.7397_{\pm 0.0034}$	$0.7013_{\pm 0.0051}$	$0.4596_{\pm 0.0041}$	$0.7420_{\pm 0.0013}$
	FedAvg	$0.6734_{\pm 0.0042}$	$0.5475_{\pm 0.0187}$	$0.6750_{\pm 0.0086}$	$0.6727_{\pm 0.0035}$	$0.5552_{\pm 0.0129}$	$0.6752_{\pm 0.0047}$	$0.6815_{\pm 0.0054}$	$0.5434_{\pm 0.0171}$	$0.6799_{\pm 0.0063}$
	FedAtt	$0.6747_{\pm 0.0056}$	$0.5457_{\pm 0.0114}$	$0.6728_{\pm 0.0067}$	$0.6797_{\pm 0.0044}$	$0.5493_{\pm 0.0134}$	$0.6700_{\pm 0.0113}$	$0.6773_{\pm 0.0008}$	$0.5561_{\pm 0.0113}$	$0.6688_{\pm 0.0079}$
	LoAdaBoost	$0.6643_{\pm 0.0019}$	$0.5711_{\pm 0.0019}$	$0.7162_{\pm 0.0015}$	$0.6664_{\pm 0.0042}$	$0.5690_{\pm 0.0032}$	$0.7198_{\pm 0.0011}$	$0.6717_{\pm 0.0030}$	$0.5439_{\pm 0.0474}$	$0.7099_{\pm 0.0256}$
	F2MF	$0.6752_{\pm 0.0053}$	$0.5456_{\pm 0.0229}$	$0.6732_{\pm 0.0042}$	$0.6680_{\pm 0.0029}$	$0.5717_{\pm 0.0033}$	$0.6685_{\pm 0.0020}$	$0.6785_{\pm 0.0059}$	$0.5550_{\pm 0.0167}$	$0.6776_{\pm 0.0038}$
	PFedRec	$0.6772_{\pm 0.0036}$	$0.5006_{\pm 0.0021}$	$0.6906_{\pm 0.0004}$	$0.6895_{\pm 0.0007}$	$0.5056_{\pm 0.0016}$	$0.6968_{\pm 0.0006}$	$0.6895_{\pm 0.0011}$	$0.5000_{\pm 0.0151}$	$0.6994_{\pm 0.0075}$
	FedCD	$0.7105_{\pm 0.0013}$	$0.5360_{\pm 0.0011}$	$0.7044_{\pm 0.0021}$	$0.7154_{\pm 0.0029}$	$0.5317_{\pm 0.0027}$	$0.7095_{\pm 0.0020}$	$0.7196_{\pm 0.0022}$	$0.5277_{\pm 0.0022}$	$0.7107_{\pm 0.0021}$
	FedCD(FedAvg)	$0.7067_{\pm 0.0004}$	$0.5363_{\pm 0.0015}$	$0.7169_{\pm 0.0019}$	$0.7119_{\pm 0.0023}$	$0.5319_{\pm 0.0025}$	$0.7201_{\pm 0.0022}$	$0.7135_{\pm 0.0022}$	$0.5306_{\pm 0.0017}$	$0.7222_{\pm 0.0010}$
	FedCD(FedAtt)	$0.7065_{\pm 0.0020}$	$0.5352_{\pm 0.0028}$	$0.7158_{\pm 0.0023}$	$0.7108_{\pm 0.0013}$	$0.5327_{\pm 0.0018}$	$0.7201_{\pm 0.0010}$	$0.7108_{\pm 0.0022}$	$0.5325_{\pm 0.0028}$	$0.7201_{\pm 0.0023}$
GUM	HPFL	$0.6740_{\pm 0.0004}$	$0.4661_{\pm 0.0011}$	$0.6729_{\pm 0.0004}$	$0.6772_{\pm 0.0013}$	$0.5473_{\pm 0.0143}$	$0.6603_{\pm 0.0115}$	$0.6826_{\pm 0.0003}$	$0.5514_{\pm 0.0003}$	$0.6612_{\pm 0.0015}$

Wilcoxon W/T/L of FedCD	vs. Base	vs. FedAvg	vs. FedAtt	vs. LoAdaBoost	vs. F2MF	vs. PFedRec
W/T/L	20/0/7	27/0/0	27/0/0	24/0/3	27/0/0	24/0/3

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Table 3. Performance of FedCD and all baseline approaches on ASSIST2012. **Bold:** the best, Underline: the runner-up. Results are reported as mean ± standard deviation over 10 independent runs. W/T/L denotes the Win/Tie/Loss results of the Wilcoxon rank-sum test at the 0.05 significance level

Base	Train/Test ratio	60%/40%			70%/30%			80%/20%		
Model	**Models**	**ACC↑**	**RMSE↓**	**AUC↑**	**ACC↑**	**RMSE↓**	**AUC↑**	**ACC↑**	**RMSE↓**	**AUC↑**
DINA	Base	$0.6924_{\pm 0.0006}$	$0.4603_{\pm 0.0002}$	$0.6522_{\pm 0.0004}$	$0.6923_{\pm 0.0007}$	$0.4607_{\pm 0.0004}$	$0.6511_{\pm 0.0011}$	$0.6929_{\pm 0.0006}$	$0.4605_{\pm 0.0003}$	$0.6523_{\pm 0.0007}$
	FedAvg	$0.6574_{\pm 0.0009}$	$0.4727_{\pm 0.0003}$	$0.6328_{\pm 0.0005}$	$0.6580_{\pm 0.0006}$	$0.4727_{\pm 0.0001}$	$0.6332_{\pm 0.0008}$	$0.6592_{\pm 0.0009}$	$0.4721_{\pm 0.0002}$	$0.6355_{\pm 0.0017}$
	FedAtt	$0.6577_{\pm 0.0005}$	$0.4726_{\pm 0.0003}$	$0.6323_{\pm 0.0009}$	$0.6592_{\pm 0.0006}$	$0.4723_{\pm 0.0003}$	$0.6344_{\pm 0.0013}$	$0.6595_{\pm 0.0007}$	$0.4721_{\pm 0.0003}$	$0.6354_{\pm 0.0017}$
	LoAdaBoost	$0.6524_{\pm 0.0007}$	$0.4736_{\pm 0.0003}$	$0.6305_{\pm 0.0007}$	$0.6527_{\pm 0.0011}$	$0.4736_{\pm 0.0004}$	$0.6324_{\pm 0.0007}$	$0.6547_{\pm 0.0010}$	$0.4729_{\pm 0.0003}$	$0.6349_{\pm 0.0007}$
	F2MF	$0.6574_{\pm 0.0007}$	$0.4725_{\pm 0.0003}$	$0.6333_{\pm 0.0005}$	$0.6583_{\pm 0.0006}$	$0.4726_{\pm 0.0002}$	$0.6334_{\pm 0.0011}$	$0.6594_{\pm 0.0009}$	$0.4720_{\pm 0.0004}$	$0.6356_{\pm 0.0011}$
	PFedRec	$0.6869_{\pm 0.0002}$	$0.4614_{\pm 0.0000}$	$0.6472_{\pm 0.0015}$	$0.6872_{\pm 0.0002}$	$0.4614_{\pm 0.0002}$	$0.6494_{\pm 0.0025}$	$0.6877_{\pm 0.0005}$	$0.4612_{\pm 0.0003}$	$0.6490_{\pm 0.0020}$
	FedCD	$0.6823_{\pm 0.0012}$	$0.4636_{\pm 0.0005}$	$0.6617_{\pm 0.0016}$	$0.6831_{\pm 0.0004}$	$0.4636_{\pm 0.0002}$	$0.6549_{\pm 0.0032}$	$0.6837_{\pm 0.0008}$	$0.4630_{\pm 0.0003}$	$0.6733_{\pm 0.0066}$
	FedCD(FedAvg)	$0.6861_{\pm 0.0006}$	$0.4612_{\pm 0.0003}$	$0.6684_{\pm 0.0030}$	$0.6865_{\pm 0.0003}$	$0.4611_{\pm 0.0002}$	$0.6723_{\pm 0.0058}$	$0.6882_{\pm 0.0004}$	$0.4606_{\pm 0.0001}$	$0.6784_{\pm 0.0036}$
	FedCD(FedAtt)	$0.6855_{\pm 0.0006}$	$0.4613_{\pm 0.0002}$	$0.6705_{\pm 0.0038}$	$0.6868_{\pm 0.0002}$	$0.4609_{\pm 0.0000}$	$0.6742_{\pm 0.0020}$	$0.6878_{\pm 0.0002}$	$0.4607_{\pm 0.0002}$	$0.6777_{\pm 0.0026}$
NCD	Base	$0.7175_{\pm 0.0003}$	$0.4360_{\pm 0.0002}$	$0.7038_{\pm 0.0006}$	$0.7181_{\pm 0.0002}$	$0.4343_{\pm 0.0012}$	$0.7102_{\pm 0.0071}$	$0.7192_{\pm 0.0002}$	$0.4341_{\pm 0.0002}$	$0.7094_{\pm 0.0003}$
	FedAvg	$0.7086_{\pm 0.0003}$	$0.4456_{\pm 0.0002}$	$0.6688_{\pm 0.0002}$	$0.7117_{\pm 0.0002}$	$0.4474_{\pm 0.0002}$	$0.6721_{\pm 0.0003}$	$0.7122_{\pm 0.0008}$	$0.4492_{\pm 0.0004}$	$0.6737_{\pm 0.0001}$
	FedAtt	$0.7095_{\pm 0.0002}$	$0.4464_{\pm 0.0002}$	$0.6680_{\pm 0.0002}$	$0.7117_{\pm 0.0002}$	$0.4483_{\pm 0.0004}$	$0.6718_{\pm 0.0002}$	$0.7096_{\pm 0.0029}$	$0.4488_{\pm 0.0022}$	$0.6730_{\pm 0.0007}$
	LoAdaBoost	$0.7113_{\pm 0.0004}$	$0.4545_{\pm 0.0003}$	$0.6645_{\pm 0.0001}$	$0.7135_{\pm 0.0003}$	$0.4540_{\pm 0.0005}$	$0.6659_{\pm 0.0002}$	$0.7100_{\pm 0.0006}$	$0.4532_{\pm 0.0012}$	$0.6663_{\pm 0.0003}$
	F2MF	$0.7118_{\pm 0.0002}$	$0.4481_{\pm 0.0005}$	$0.6701_{\pm 0.0001}$	$0.7102_{\pm 0.0002}$	$0.4457_{\pm 0.0002}$	$0.6725_{\pm 0.0000}$	$0.7131_{\pm 0.0004}$	$0.4467_{\pm 0.0007}$	$0.6741_{\pm 0.0003}$
	PFedRec	$0.7205_{\pm 0.0001}$	$0.4363_{\pm 0.0001}$	$0.6965_{\pm 0.0000}$	$0.7214_{\pm 0.0003}$	$0.4357_{\pm 0.0002}$	$0.7008_{\pm 0.0000}$	$0.7224_{\pm 0.0001}$	$0.4347_{\pm 0.0001}$	$0.7045_{\pm 0.0000}$
	FedCD	$0.7177_{\pm 0.0003}$	$0.4491_{\pm 0.0004}$	$0.7014_{\pm 0.0005}$	$0.7181_{\pm 0.0007}$	$0.4471_{\pm 0.0009}$	$0.7045_{\pm 0.0004}$	$0.7221_{\pm 0.0003}$	$0.4439_{\pm 0.0013}$	$0.7092_{\pm 0.0016}$
	FedCD(FedAvg)	$0.7227_{\pm 0.0003}$	$0.4370_{\pm 0.0003}$	$0.7022_{\pm 0.0006}$	$0.7240_{\pm 0.0004}$	$0.4360_{\pm 0.0002}$	$0.7056_{\pm 0.0005}$	$0.7260_{\pm 0.0003}$	$0.4352_{\pm 0.0003}$	$0.7081_{\pm 0.0004}$
	FedCD(FedAtt)	$0.7225_{\pm 0.0003}$	$0.4370_{\pm 0.0002}$	$0.7023_{\pm 0.0008}$	$0.7246_{\pm 0.0005}$	$0.4362_{\pm 0.0002}$	$0.7056_{\pm 0.0010}$	$0.7258_{\pm 0.0003}$	$0.4350_{\pm 0.0002}$	$0.7082_{\pm 0.0006}$
KaNCD	Base	$0.7163_{\pm 0.0030}$	$0.4473_{\pm 0.0154}$	$0.7158_{\pm 0.0026}$	$0.7125_{\pm 0.0017}$	$0.4506_{\pm 0.0174}$	$0.7134_{\pm 0.0037}$	$0.7219_{\pm 0.0022}$	$0.4461_{\pm 0.0099}$	$0.7182_{\pm 0.0033}$
	FedAvg	$0.6807_{\pm 0.0195}$	$0.5083_{\pm 0.0312}$	$0.6494_{\pm 0.0061}$	$0.6693_{\pm 0.0192}$	$0.5152_{\pm 0.0338}$	$0.6534_{\pm 0.0041}$	$0.6616_{\pm 0.0143}$	$0.5252_{\pm 0.0323}$	$0.6515_{\pm 0.0069}$
	FedAtt	$0.6874_{\pm 0.0175}$	$0.4976_{\pm 0.0056}$	$0.6474_{\pm 0.0039}$	$0.6591_{\pm 0.0056}$	$0.5552_{\pm 0.0131}$	$0.6593_{\pm 0.0020}$	$0.6625_{\pm 0.0141}$	$0.5451_{\pm 0.0286}$	$0.6573_{\pm 0.0028}$
	LoAdaBoost	$0.6728_{\pm 0.0075}$	$0.5171_{\pm 0.0119}$	$0.6581_{\pm 0.0015}$	$0.6747_{\pm 0.0216}$	$0.5059_{\pm 0.0262}$	$0.6527_{\pm 0.0021}$	$0.6729_{\pm 0.0166}$	$0.5191_{\pm 0.0177}$	$0.6555_{\pm 0.0040}$
	F2MF	$0.6597_{\pm 0.0206}$	$0.5515_{\pm 0.0448}$	$0.6573_{\pm 0.0056}$	$0.6739_{\pm 0.0240}$	$0.5273_{\pm 0.0483}$	$0.6561_{\pm 0.0068}$	$0.6734_{\pm 0.0208}$	$0.5252_{\pm 0.0360}$	$0.6547_{\pm 0.0053}$
	PFedRec	$0.6799_{\pm 0.0017}$	$0.5444_{\pm 0.0359}$	$0.6534_{\pm 0.0195}$	$0.6950_{\pm 0.0065}$	$0.4864_{\pm 0.0011}$	$0.6860_{\pm 0.0002}$	$0.6768_{\pm 0.0065}$	$0.5348_{\pm 0.0425}$	$0.6630_{\pm 0.0306}$
	FedCD	$0.7085_{\pm 0.0049}$	$0.5350_{\pm 0.0014}$	$0.6631_{\pm 0.0142}$	$0.7102_{\pm 0.0026}$	$0.5351_{\pm 0.0022}$	$0.6596_{\pm 0.0040}$	$0.7121_{\pm 0.0016}$	$0.5335_{\pm 0.0013}$	$0.6639_{\pm 0.0022}$
	FedCD(FedAvg)	$0.7006_{\pm 0.0017}$	$0.5352_{\pm 0.0016}$	$0.6849_{\pm 0.0010}$	$0.7015_{\pm 0.0011}$	$0.5357_{\pm 0.0036}$	$0.6841_{\pm 0.0014}$	$0.7038_{\pm 0.0009}$	$0.5321_{\pm 0.0008}$	$0.6865_{\pm 0.0017}$
	FedCD(FedAtt)	$0.7025_{\pm 0.0017}$	$0.5336_{\pm 0.0018}$	$0.6853_{\pm 0.0005}$	$0.7021_{\pm 0.0020}$	$0.5353_{\pm 0.0041}$	$0.6836_{\pm 0.0026}$	$0.7045_{\pm 0.0021}$	$0.5317_{\pm 0.0016}$	$0.6861_{\pm 0.0012}$
GUM	HPFL	$0.7023_{\pm 0.0001}$	$0.4518_{\pm 0.0033}$	$0.6556_{\pm 0.0014}$	$0.7013_{\pm 0.0002}$	$0.4514_{\pm 0.0003}$	$0.6578_{\pm 0.0003}$	$0.7019_{\pm 0.0003}$	$0.4643_{\pm 0.0226}$	$0.6619_{\pm 0.0057}$

Wilcoxon W/T/L of FedCD	**vs. Base**	**vs. FedAvg**	**vs. FedAtt**	**vs. LoAdaBoost**	**vs. F2MF**	**vs. PFedRec**
W/T/L	6/1/20	23/2/2	25/0/2	24/0/3	23/2/2	13/0/14

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- **NCD** is one of the latest CDMs that introduces deep learning. It employs a neural network instead of a manually designed prediction function to predict students' responses.
- **KaNCD** is a knowledge-association-based extension of NCD, which employs higher-order latent representations of students, exercises, and knowledge concepts to model latent knowledge associations [49]. Different from NCD, its student proficiency and exercise difficulty representations are jointly constructed through interactions with knowledge-concept representations, resulting in stronger coupling among different model components.

The FL approaches compared in this paper can be categorized into three groups. The first group comprises general federated learning approaches, such as FedAvg, FedAtt, and LoAdaBoost. The second group includes one FL approach based on personalization techniques, i.e., PFedRec, and one FL approach designed to address fairness issues, i.e., F2MF. The final group consists of the proposed FedCD and its variants. Additionally, there is an FL approach for CD, termed HPFL, which is the first work to apply FL techniques to CD. Detailed descriptions of these approaches are provided below.

- **FedAvg:** This is a classical federated learning algorithm that averages local model updates from participating clients to create a global model. It is widely used due to its simplicity and scalability across various applications.
- **FedAtt:** This approach extends FedAvg by incorporating an attention mechanism to assign adaptive weights to client updates based on their importance, thereby enhancing the robustness and fairness of the global model.
- **LoAdaBoost:** This is a federated boosting approach that iteratively adjusts the weights of clients, emphasizing those with lower performance to achieve better accuracy and fairness in the global model.
- **Fairness-aware Federated Matrix Factorization (F2MF):** This approach aims to ensure fairness across different user groups in federated learning. To achieve this, it facilitates the communication of group-level statistics during the federated optimization process. Additionally, it incorporates differential privacy techniques to protect users' group information, particularly in scenarios where privacy protection is essential.
- **Personalized Federated Recommendation (PFedRec):** This approach introduces a personalized federated learning framework tailored for recommendation tasks, combining global knowledge aggregation with personalized model adaptation to enhance recommendation accuracy for individual users.
- **Hierarchical Personalized Federated Learning (HPFL):** This approach designs a local user model GUM that applies different aggregation strategies for private and public components. Despite variations in settings across different clients, the model ensures both similarity and personalization in its components.

The descriptions of the variants of FedCD have been presented previously.

There are several variants of FedCD designed to validate the effectiveness of its different components, described as follows:

- *FedCD(FedAvg)*: Similar to FedCD, but uses average aggregation from FedAvg for the exercise embedding parameters.
- *FedCD(FedAvg)-s.*: Same as *FedCD(FedAvg)*, but also applies average aggregation for the student embedding parameters.
- *FedCD(FedAvg)-p.d.p.*: FedCD without the parameter decoupling-based personalization strategy, i.e., sharing the diagnostic function-related parameters θ_t^I across all clients, and using average aggregation for both exercise embedding parameters and diagnostic function-related parameters.
- *FedCD(FedAtt)*: Similar to FedCD, but employs attention-based aggregation from FedAtt for the exercise embedding parameters.

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- *FedCD(FedAtt)-s.*: Same as *FedCD(FedAtt)*, but also applies attention-based aggregation for the student embedding parameters.
- *FedCD(FedAtt)-p.d.p.*: Same as *FedCD(FedAvg)-p.d.p.*, but uses attention-based aggregation for the exercise embedding parameters and the diagnostic function parameters.

5.1.3 Evaluation Metrics. According to the experiences of previous research [6, 12, 48], three metrics were adopted to measure the performance of all CDMs [59], including *area under the curve* (AUC), *accuracy* (ACC), and *root mean square error* (RMSE).

Since client fairness is a central objective of FedCD, we explicitly evaluate the distribution of diagnosis performance across clients using three complementary fairness metrics: group fairness disparity (GF), client accuracy standard deviation (STD_{ACC}), and worst-client accuracy (WACC). Let A_t denote the ACC of the t-th client and let

$$\bar{A} = \frac{1}{T}\sum_{t=1}^{T} A_t \tag{13}$$

denote the average client accuracy (Avg. ACC). Avg. ACC is additionally reported as an overall diagnostic utility metric, but is not itself used as evidence of client fairness.

First, following [71], clients are divided into two groups according to the mean client ACC, i.e., $G_0 = \{t|A_t < \bar{A}\}$ and $G_1 = \{t|A_t \geq \bar{A}\}$. We use group fairness disparity (GF) to measure the performance difference between the two groups:

$$GF = \left| \frac{1}{|G_0|}\sum_{t\in G_0} A_t - \frac{1}{|G_1|}\sum_{t\in G_1} A_t \right|. \tag{14}$$

Note that GF is a disparity measure for evaluating client fairness rather than a fairness score itself. A smaller GF indicates a smaller performance gap between the two client groups and thus better client fairness.

Second, we use the standard deviation of client accuracy,

$$STD_{ACC} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(A_t - \bar{A})^2}, \tag{15}$$

to measure the overall dispersion of client performance, where a smaller value indicates more balanced diagnosis performance across clients. Third, worst-client accuracy (WACC) is defined as

$$WACC = \min_t A_t, \tag{16}$$

which measures the diagnostic utility received by the worst-performing client, and a larger value is desirable. Therefore, GF, STD_{ACC}, and WACC characterize client fairness from the perspectives of group-level disparity, overall client-performance dispersion, and worst-client performance, respectively. Smaller GF and STD_{ACC} and larger WACC indicate better client fairness, whereas Avg. ACC is reported separately to assess the overall diagnostic utility.

All compared methods were independently run 10 times, and their performance is reported in terms of mean and standard deviation. A larger ACC/AUC and a smaller RMSE indicate better diagnosis performance. To examine whether the performance differences are statistically significant, the Wilcoxon rank-sum test [7] with a significance level of 0.05 was conducted between FedCD and each compared approach for each combination of base CDM, train/test ratio, and evaluation metric. The statistical results are summarized as Win/Tie/Loss (W/T/L), where Win and Loss indicate that FedCD performs significantly better and worse than the compared approach, respectively, according to the desirable direction of each metric, while Tie indicates that no statistically significant difference is observed.

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5.1.4 Parameter Settings. For the proposed FedCD, its dimension D was set to the number of concepts, i.e., K, γ was set to 0.1, the maximal number of rounds P was set to 100, the number of local training epochs Num was set to 5, the Adam optimizer was employed for local model training with a learning rate of 0.001 and batch size of 128. For comprehensive comparisons, three splitting settings (60%/40%, 70%/30%, and 80%/20%) were adopted to get training and testing datasets. All compared CDMs followed the settings in their original papers, and FedAvg, FedAtt, and LoAdaBoost hold the same settings as FedCD. All models were implemented by PyTorch, and all experiments were conducted on an NVIDIA 3090 GPU. The source code of FedCD is at https://github.com/NullPointer233/FedCD.

5.2 Overall Performance Comparison (RQ1)

To answer **RQ1**, FedCD was compared with *Base*, FedAvg, FedAtt, LoAdaBoost, F2MF, and PFedRec under DINA, NCD, and KaNCD on three datasets: ASSIST2009, ASSIST2012, and SLP-Math, where *Base* denotes the centralized training paradigm. Tables 2 and 3 present their performance on ASSIST2009 and ASSIST2012 regarding AUC, ACC, and RMSE obtained under three splitting settings. Only the best result among the comparisons is shown in bold, and the second-best result is underlined.

As we can observe, although the centralized base models remain competitive or perform better in some settings, FedCD exhibits better overall performance than the compared FL approaches in most cases. To further examine whether these performance differences are statistically meaningful, the lower parts of Tables 2 and 3 report the Win/Tie/Loss results obtained by the Wilcoxon rank-sum test at the 0.05 significance level. Each table contains 27 pairwise comparisons between FedCD and each compared approach, considering three base CDMs, three train/test ratios, and three evaluation metrics. The statistical results show that FedCD obtains more significant wins than losses against most of the compared FL approaches, further supporting the overall performance advantage observed from the mean results. Meanwhile, the reported ties explicitly distinguish statistically insignificant differences from significant improvements, avoiding conclusions based only on small numerical differences in the mean performance. The NCD-based FedCD also achieves better diagnosis performance than HPFL across the considered metrics, further demonstrating the effectiveness of FedCD in federated cognitive diagnosis. Additionally, compared to the two variants, the overall performance of FedCD is slightly better than *FedCD(Avg)* and *FedCD(Att)*, confirming the effectiveness of the utilized fairness-aware aggregation mechanism.

To further evaluate the effectiveness of FedCD compared to other FL approaches under different CDMs, the overall comparison results on the SLP-Math dataset are summarized in Table 4, which used the same settings as the experiments on ASSIST2009 and ASSIST2012.

As shown in Table 4, FedCD and its two variants, *FedCD(FedAvg)* and *FedCD(FedAtt)*, generally achieve better overall performance than the compared FL approaches. The Wilcoxon W/T/L results in the lower part of Table 4 further support this observation, with FedCD obtaining more significant wins than losses against the compared FL approaches. Furthermore, the performance of FedCD based on NCD and DINA even surpasses that of the centralized model, demonstrating that the proposed aggregation strategy can enhance the overall performance of the base models.

It is also worth noting that the performance improvement of FedCD under KaNCD is less consistent than that under DINA and NCD. We attribute this mainly to the stronger coupling among different representations in KaNCD. Specifically, KaNCD models latent knowledge associations using higher-order representations of students, exercises, and knowledge concepts, where student proficiency and exercise difficulty are jointly determined by their interactions with knowledge-concept representations. Such a more complex representation mechanism is more sensitive to the fragmented local data in the federated setting, since each client has substantially fewer response records than in centralized training. This observation is also reflected by the results of other FL approaches: compared with the centralized KaNCD, FedAvg, FedAtt, F2MF, and other federated baselines

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Table 4. Performance of FedCD and all baseline approaches on SLP-Math. **Bold:** the best, Underline: the runner-up. Results are reported as mean ± standard deviation over 10 independent runs. W/T/L denotes the Win/Tie/Loss results of the Wilcoxon rank-sum test at the 0.05 significance level

Base	Train/Test ratio	60%/40%			70%/30%			80%/20%		
Model	Models	ACC↑	RMSE↓	AUC↑	ACC↑	RMSE↓	AUC↑	ACC↑	RMSE↓	AUC↑
DINA	Base	$0.6470_{\pm 0.0010}$	$0.4781_{\pm 0.0004}$	$0.7011_{\pm 0.0045}$	$0.6498_{\pm 0.0014}$	$0.4766_{\pm 0.0003}$	$0.7071_{\pm 0.0020}$	$0.6486_{\pm 0.0009}$	$0.4776_{\pm 0.0001}$	$0.7043_{\pm 0.0023}$
	FedAvg	$0.6354_{\pm 0.0014}$	$0.4783_{\pm 0.0005}$	$0.6962_{\pm 0.0028}$	$0.6388_{\pm 0.0026}$	$0.4777_{\pm 0.0007}$	$0.7022_{\pm 0.0019}$	$0.6427_{\pm 0.0024}$	$0.4780_{\pm 0.0005}$	$0.7059_{\pm 0.0019}$
	FedAtt	$0.6446_{\pm 0.0015}$	$0.4759_{\pm 0.0006}$	$0.6974_{\pm 0.0018}$	$0.6472_{\pm 0.0021}$	$0.4745_{\pm 0.0006}$	$0.7028_{\pm 0.0022}$	$0.6502_{\pm 0.0009}$	$0.4743_{\pm 0.0005}$	$0.7068_{\pm 0.0033}$
	LoAdaBoost	$0.6412_{\pm 0.0012}$	$0.4771_{\pm 0.0006}$	$0.7039_{\pm 0.0040}$	$0.6348_{\pm 0.0026}$	$0.4785_{\pm 0.0003}$	$0.7055_{\pm 0.0051}$	$0.6418_{\pm 0.0036}$	$0.4784_{\pm 0.0004}$	$0.7135_{\pm 0.0035}$
	F2MF	$0.6381_{\pm 0.0015}$	$0.4781_{\pm 0.0004}$	$0.6964_{\pm 0.0024}$	$0.6364_{\pm 0.0014}$	$0.4786_{\pm 0.0005}$	$0.6969_{\pm 0.0013}$	$0.6446_{\pm 0.0010}$	$0.4774_{\pm 0.0003}$	$0.7071_{\pm 0.0023}$
	PFedRec	$0.6399_{\pm 0.0016}$	$0.4794_{\pm 0.0006}$	$0.7004_{\pm 0.0073}$	$0.6382_{\pm 0.0030}$	$0.4811_{\pm 0.0011}$	$0.7100_{\pm 0.0088}$	$0.6388_{\pm 0.0023}$	$0.4803_{\pm 0.0008}$	$0.7008_{\pm 0.0083}$
	FedCD	$0.6800_{\pm 0.0012}$	$0.4664_{\pm 0.0004}$	$0.7661_{\pm 0.0015}$	$0.6868_{\pm 0.0007}$	$0.4624_{\pm 0.0006}$	$0.7546_{\pm 0.0017}$	$0.6863_{\pm 0.0011}$	$0.4640_{\pm 0.0005}$	$0.7604_{\pm 0.0005}$
	FedCD(FedAvg)	$0.6786_{\pm 0.0003}$	$0.4652_{\pm 0.0007}$	$0.7566_{\pm 0.0050}$	$0.6800_{\pm 0.0016}$	$0.4628_{\pm 0.0006}$	$0.7539_{\pm 0.0013}$	$0.6819_{\pm 0.0018}$	$0.4636_{\pm 0.0001}$	$0.7596_{\pm 0.0026}$
	FedCD(FedAtt)	$0.6796_{\pm 0.0017}$	$0.4652_{\pm 0.0007}$	$0.7532_{\pm 0.0049}$	$0.6781_{\pm 0.0022}$	$0.4636_{\pm 0.0010}$	$0.7508_{\pm 0.0012}$	$0.6818_{\pm 0.0017}$	$0.4636_{\pm 0.0003}$	$0.7562_{\pm 0.0058}$
NCD	Base	$0.7227_{\pm 0.0008}$	$0.4512_{\pm 0.0077}$	$0.7880_{\pm 0.0013}$	$0.7296_{\pm 0.0016}$	$0.4464_{\pm 0.0026}$	$0.7972_{\pm 0.0017}$	$0.7326_{\pm 0.0024}$	$0.4474_{\pm 0.0018}$	$0.7989_{\pm 0.0009}$
	FedAvg	$0.6916_{\pm 0.0015}$	$0.4535_{\pm 0.0006}$	$0.7591_{\pm 0.0013}$	$0.6990_{\pm 0.0013}$	$0.4524_{\pm 0.0005}$	$0.7650_{\pm 0.0002}$	$0.7010_{\pm 0.0024}$	$0.4515_{\pm 0.0008}$	$0.7688_{\pm 0.0017}$
	FedAtt	$0.6891_{\pm 0.0010}$	$0.4580_{\pm 0.0004}$	$0.7532_{\pm 0.0004}$	$0.6965_{\pm 0.0017}$	$0.4566_{\pm 0.0006}$	$0.7591_{\pm 0.0008}$	$0.7000_{\pm 0.0009}$	$0.4543_{\pm 0.0006}$	$0.7641_{\pm 0.0005}$
	LoAdaBoost	$0.6973_{\pm 0.0006}$	$0.4642_{\pm 0.0005}$	$0.7693_{\pm 0.0006}$	$0.7062_{\pm 0.0011}$	$0.4546_{\pm 0.0009}$	$0.7761_{\pm 0.0006}$	$0.7050_{\pm 0.0011}$	$0.4546_{\pm 0.0006}$	$0.7771_{\pm 0.0004}$
	F2MF	$0.6925_{\pm 0.0007}$	$0.4522_{\pm 0.0005}$	$0.7603_{\pm 0.0009}$	$0.7002_{\pm 0.0015}$	$0.4515_{\pm 0.0004}$	$0.7664_{\pm 0.0004}$	$0.7015_{\pm 0.0007}$	$0.4498_{\pm 0.0008}$	$0.7706_{\pm 0.0005}$
	PFedRec	$0.6778_{\pm 0.0003}$	$0.4686_{\pm 0.0011}$	$0.7659_{\pm 0.0001}$	$0.6734_{\pm 0.0013}$	$0.4693_{\pm 0.0011}$	$0.7764_{\pm 0.0006}$	$0.6714_{\pm 0.0023}$	$0.4693_{\pm 0.0051}$	$0.7746_{\pm 0.0037}$
	FedCD	$0.7283_{\pm 0.0009}$	$0.4442_{\pm 0.0023}$	$0.7788_{\pm 0.0011}$	$0.7347_{\pm 0.0006}$	$0.4402_{\pm 0.0029}$	$0.7909_{\pm 0.0021}$	$0.7374_{\pm 0.0010}$	$0.4335_{\pm 0.0013}$	$0.7934_{\pm 0.0023}$
	FedCD(FedAvg)	$0.7236_{\pm 0.0007}$	$0.4490_{\pm 0.0011}$	$0.7764_{\pm 0.0013}$	$0.7335_{\pm 0.0014}$	$0.4395_{\pm 0.0012}$	$0.7898_{\pm 0.0015}$	$0.7349_{\pm 0.0005}$	$0.4385_{\pm 0.0019}$	$0.7913_{\pm 0.0014}$
	FedCD(FedAtt)	$0.7227_{\pm 0.0007}$	$0.4490_{\pm 0.0015}$	$0.7750_{\pm 0.0014}$	$0.7342_{\pm 0.0007}$	$0.4404_{\pm 0.0017}$	$0.7890_{\pm 0.0002}$	$0.7339_{\pm 0.0007}$	$0.4383_{\pm 0.0019}$	$0.7927_{\pm 0.0008}$
KaNCD	Base	$0.7374_{\pm 0.0018}$	$0.4281_{\pm 0.0049}$	$0.8081_{\pm 0.0011}$	$0.7431_{\pm 0.0009}$	$0.4247_{\pm 0.0015}$	$0.8139_{\pm 0.0013}$	$0.7419_{\pm 0.0020}$	$0.4264_{\pm 0.0026}$	$0.8134_{\pm 0.0011}$
	FedAvg	$0.7071_{\pm 0.0013}$	$0.5369_{\pm 0.0014}$	$0.7634_{\pm 0.0012}$	$0.7085_{\pm 0.0031}$	$0.5350_{\pm 0.0018}$	$0.7641_{\pm 0.0028}$	$0.7045_{\pm 0.0018}$	$0.5391_{\pm 0.0015}$	$0.7660_{\pm 0.0031}$
	FedAtt	$0.7082_{\pm 0.0014}$	$0.5356_{\pm 0.0011}$	$0.7619_{\pm 0.0021}$	$0.7120_{\pm 0.0022}$	$0.5327_{\pm 0.0026}$	$0.7630_{\pm 0.0015}$	$0.7093_{\pm 0.0027}$	$0.5349_{\pm 0.0019}$	$0.7636_{\pm 0.0020}$
	LoAdaBoost	$0.7179_{\pm 0.0017}$	$0.5260_{\pm 0.0010}$	$0.7749_{\pm 0.0008}$	$0.7171_{\pm 0.0014}$	$0.5261_{\pm 0.0013}$	$0.7787_{\pm 0.0015}$	$0.7131_{\pm 0.0011}$	$0.5299_{\pm 0.0011}$	$0.7774_{\pm 0.0021}$
	F2MF	$0.7109_{\pm 0.0012}$	$0.5341_{\pm 0.0006}$	$0.7649_{\pm 0.0015}$	$0.7098_{\pm 0.0019}$	$0.5349_{\pm 0.0021}$	$0.7676_{\pm 0.0020}$	$0.7085_{\pm 0.0034}$	$0.5351_{\pm 0.0024}$	$0.7673_{\pm 0.0041}$
	PFedRec	$0.6939_{\pm 0.0019}$	$0.5461_{\pm 0.0015}$	$0.7614_{\pm 0.0008}$	$0.6821_{\pm 0.0120}$	$0.5575_{\pm 0.0115}$	$0.7695_{\pm 0.0015}$	$0.6833_{\pm 0.0082}$	$0.5581_{\pm 0.0076}$	$0.7711_{\pm 0.0006}$
	FedCD	$0.7255_{\pm 0.0028}$	$0.5211_{\pm 0.0029}$	$0.7556_{\pm 0.0043}$	$0.7256_{\pm 0.0019}$	$0.5211_{\pm 0.0018}$	$0.7536_{\pm 0.0019}$	$0.7244_{\pm 0.0022}$	$0.5222_{\pm 0.0021}$	$0.7527_{\pm 0.0030}$
	FedCD(FedAvg)	$0.7226_{\pm 0.0014}$	$0.5242_{\pm 0.0019}$	$0.7677_{\pm 0.0031}$	$0.7245_{\pm 0.0026}$	$0.5215_{\pm 0.0021}$	$0.7693_{\pm 0.0042}$	$0.7227_{\pm 0.0016}$	$0.5228_{\pm 0.0020}$	$0.7689_{\pm 0.0068}$
	FedCD(FedAtt)	$0.7213_{\pm 0.0020}$	$0.5214_{\pm 0.0044}$	$0.7683_{\pm 0.0091}$	$0.7254_{\pm 0.0011}$	$0.5203_{\pm 0.0018}$	$0.7648_{\pm 0.0067}$	$0.7240_{\pm 0.0013}$	$0.5219_{\pm 0.0016}$	$0.7676_{\pm 0.0059}$
GUM	HPFL	$0.6864_{\pm 0.0021}$	$0.5195_{\pm 0.0020}$	$0.7401_{\pm 0.0027}$	$0.6978_{\pm 0.0016}$	$0.5142_{\pm 0.0016}$	$0.7502_{\pm 0.0011}$	$0.7026_{\pm 0.0005}$	$0.5123_{\pm 0.0006}$	$0.7505_{\pm 0.0015}$

Wilcoxon W/T/L of FedCD	vs. Base	vs. FedAvg	vs. FedAtt	vs. LoAdaBoost	vs. F2MF	vs. PFedRec
W/T/L	15/0/12	24/0/3	24/0/3	24/0/3	24/0/3	24/0/3

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Table 5. Performance of FedCD and its six variants on two datasets based on NCD under the ratio of 80%/20%

Models	ASSIST2009			ASSIST2012		
	ACC↑	RMSE↓	AUC↑	ACC↑	RMSE↓	AUC↑
FedAvg	67.99	46.01	67.20	71.13	44.71	67.23
FedCD(Avg)-p.d.p.	68.49	46.50	71.87	71.56	44.99	70.83
FedCD(Avg)	70.29	44.63	73.15	71.76	44.08	71.44
FedAtt	67.61	46.94	66.55	71.13	44.89	67.16
FedCD(Att)-p.d.p.	68.01	46.89	71.33	71.46	44.98	70.85
FedCD(Att)	70.30	44.69	73.62	71.58	44.39	71.48
FedCD	71.43	44.09	74.15	71.94	44.70	70.43

generally exhibit larger performance degradation under KaNCD than under DINA and NCD. Therefore, the relatively weaker performance is not specific to the fairness-aware aggregation in FedCD, but partly originates from the higher sensitivity of KaNCD to federated training.

Moreover, the current FedCD adopts a unified parameter-decoupling rule for different CDMs, where exercise-related representations are globally shared while student-side and diagnostic-function-related information is locally personalized. This decomposition is relatively natural for CDMs such as NCD, whose student embedding, exercise embedding, and diagnostic function can be more clearly separated. In KaNCD, however, exercise representations interact more strongly with student and knowledge-concept representations. Consequently, globally aggregating only the exercise-side information cannot fully transfer the coupled student–exercise–concept association knowledge across clients, which may limit the benefit of FedCD under KaNCD. Nevertheless, FedCD still achieves competitive performance and generally improves over conventional FL approaches under KaNCD, demonstrating that the framework remains applicable to this more complex CDM. This observation also reveals a limitation of the current unified decoupling strategy: CDM-specific or architecture-aware parameter decoupling may further improve FedCD when applied to models with strongly coupled representation structures, which will be investigated in future work.

5.3 Effectiveness of Parameter Decoupling based Personalization Strategy (RQ2)

5.3.1 Parameter Decoupling Effectiveness. The comparisons between FedCD and *FedCD(Avg)* as well as *FedCD(Att)* in Tables 2 and 3 have demonstrated the effectiveness of the fairness-aware aggregation mechanism in the proposed parameter decoupling-based personalization strategy to some extent.

To further explore the effectiveness of this strategy, we created two variants of the proposed FedCD, denoted as *FedCD(Avg)-p.d.p.*, and *FedCD(Att)-p.d.p.*, with detailed descriptions summarized in **Section 5.1.2.** Table 5 presents the performance of FedCD and its six variants on two datasets using NCD as the base model. As shown in Table 5, the comparisons between *FedCD(Avg)* and *FedCD(Avg)-p.d.p.* confirm the effectiveness of the role of decoupling parameters for personalization in the strategy. The same conclusion can be drawn from the results of the *FedCD(Att)* series.

5.3.2 Effectiveness of Globally Shared Exercise Embeddings. To further investigate the effectiveness of globally sharing exercise embeddings, we construct a variant of FedCD, termed *FedCD_per*, in which the exercise embeddings are also kept locally personalized and are no longer aggregated across clients. The other components

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Table 6. ACC comparison between FedCD and *FedCD_per* on five selected low-performing schools of ASSIST2009 under the NCD base model

Models / ACC↑	School 4	School 7	School 8	School 10	School 14	Average ACC↑
FedCD_per	0.5883	0.6051	0.6974	0.6720	0.5960	0.6318
FedCD	**0.6338**	**0.6351**	**0.7367**	**0.7007**	**0.6582**	**0.6729**

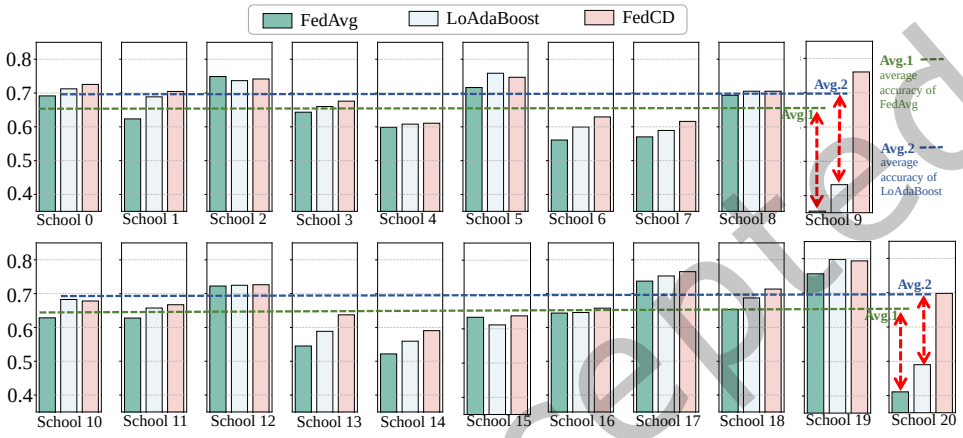


Fig. 3. Each client's ACC values of FedCD, FedAvg, and LoAdaBoost on ASSIST2009 under NCD.

and experimental settings remain the same as those of FedCD. Therefore, this variant removes the cross-client information-sharing channel provided by the globally shared exercise embeddings.

Since the client fairness issue mainly concerns clients with relatively poor diagnosis performance, we randomly select five schools from the below-average-performing clients and conduct the experiment on ASSIST2009 using NCD as the base model. Table 6 reports their client-level ACC values.

As shown in Table 6, FedCD consistently outperforms FedCD_per on all selected low-performing schools. This result indicates that keeping exercise embeddings completely local limits these clients to learning exercise representations only from their own response data. In contrast, globally shared exercise embeddings enable them to exploit exercise-related knowledge learned across schools, thereby improving their diagnosis performance. This observation provides direct evidence for the effectiveness of globally sharing exercise embeddings in FedCD. The broader effect of this design on client fairness is further evaluated using multiple fairness metrics in **RQ3**.

5.4 Client Fairness Evaluation (RQ3)

5.4.1 Client-wise Performance Analysis. To answer **RQ3**, Fig. 3 shows the ACC values for each client/school obtained by FedAvg, LoAdaBoost, and FedCD under the NCD base model on ASSIST2009. It is evident that FedCD maintains a certain level of accuracy advantage over FedAvg and LoAdaBoost for each client/school, with this advantage being particularly significant in Schools 9 and 20. For a deeper insight into the results, Table 7 summarizes the statistics of each client's student learning data in the upper part and presents each client's ACC for FedAvg, LoAdaBoost, and FedCD on ASSIST2009 in the lower part. It can be inferred that the significantly lower ACC values of FedAvg and LoAdaBoost in Schools 9 and 20 are not influenced by the number of students

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Table 7. Each client's statistics and predicted ACC

Statistics / ACC↑	School 0	1	...	9	...	19	20
# of students	205	306		150		74	65
# of wrongs	7176	7767		4003		259	536
# of corrects	15827	10408	...	1642	...	818	327
# of logs	23003	18175		5645		1077	863
Correct rate	0.688	0.573		0.291		0.759	0.379
FedAvg	69.15	62.35		35.46		75.60	41.09
LoAdaBoost	71.24	68.87	...	43.17	...	79.76	49.01
FedCD	72.55	70.47		76.05		79.37	69.80

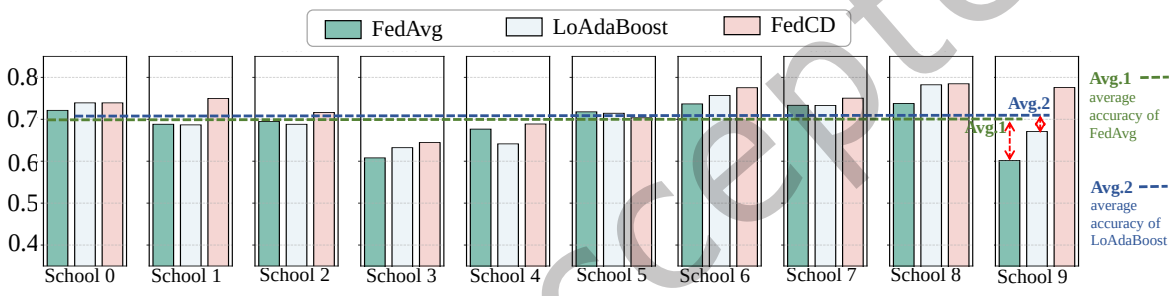


Fig. 4. Each client's ACC values of FedCD, FedAvg, and LoAdaBoost on the SLP-Math dataset under the NCD base model.

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or the number of logs contained in each client. The unusually low ACC values for these two clients compared to others may be attributed to their lower correct response rates.

Therefore, the poor diagnosis performance of Schools 9 and 20 is not simply associated with smaller client data sizes, but is closely related to their different response-data distributions. FedCD substantially improves the ACC of these poorly performing clients and makes their diagnosis performance closer to that of the other schools. These client-wise observations provide intuitive evidence that FedCD can improve diagnostic utility for underperforming clients. A formal evaluation of whether such improvements reduce the overall performance disparity among clients is presented in the following quantitative fairness evaluation.

To further investigate the effectiveness of FedCD in addressing the client fairness issue, we conducted additional experiments using FedAvg, LoAdaBoost, and FedCD based on NCD on the SLP-Math dataset. Fig 4 illustrates the ACC values of the three approaches across different schools.

The results show that FedCD consistently improves the diagnostic ACC of clients with relatively poor performance, providing further client-level evidence on another dataset. Notably, the improvement is most pronounced for School 9, which contains a relatively large proportion of incorrect responses and thus exhibits a local response distribution that differs substantially from those of the other schools. FedCD substantially improves the ACC of this client, providing further evidence of its ability to accommodate heterogeneous client response distributions.

Table 8. Each client's ACC values of FedCD, FedAvg, and LoAdaBoost on the ASSIST2012 dataset under the KaNCD base model

Models / ACC↑	School 0	School 1	School 2	School 3	School 4	School 5	School 6	School 7	School 8	School 9
FedAvg	0.5810	0.6551	0.6143	0.6799	0.6343	0.6490	0.7002	0.5504	0.6422	0.7111
LoAdaBoost	0.6913	0.6543	0.6742	0.6999	0.6633	0.6357	0.6829	0.4822	0.6238	0.6659
FedCD	**0.7557**	**0.6983**	**0.7213**	**0.7122**	**0.7204**	**0.6642**	**0.7273**	**0.6217**	**0.6654**	**0.7494**

The following Table 8 shows additional experiments on ASSIST2012. We randomly selected 10 schools for the experimental results. The results are consistent with the client-wise observations on ASSIST2009 and SLP-Math, showing that FedCD improves the diagnosis performance of the selected clients on ASSIST2012. These results serve as complementary client-level evidence, while the overall client-performance disparity is quantitatively evaluated below using explicit fairness metrics.

Taken together, these client-wise comparisons show that FedCD can substantially improve diagnosis utility for several underperforming clients. However, client-wise ACC improvements alone do not constitute a formal evaluation of client fairness. Therefore, we further quantify the overall performance disparity across clients using explicit fairness metrics in the following subsection.

Table 9. Quantitative client-fairness comparison on three datasets under the NCD base model and the 80%/20% splitting setting. Avg. ACC denotes the average client-level ACC and is reported as an overall utility metric. GF, STD_{ACC}, and WACC evaluate client fairness from complementary perspectives. ↑ and ↓ indicate that larger and smaller values are better, respectively

Models	ASSIST2009				ASSIST2012				SLP-Math			
	Avg. ACC↑	GF↓	STD_{ACC}↓	WACC↑	Avg. ACC↑	GF↓	STD_{ACC}↓	WACC↑	Avg. ACC↑	GF↓	STD_{ACC}↓	WACC↑
FedAvg	0.680	0.167	0.112	0.355	0.712	0.121	0.081	0.550	0.701	0.146	0.097	0.506
FedAtt	0.676	0.170	0.090	0.360	0.710	0.120	0.062	0.480	0.700	0.135	0.070	0.550
LoAdaBoost	0.698	0.142	0.093	0.432	0.710	0.116	0.079	0.563	0.705	0.131	0.088	0.536
F2MF	0.682	0.158	0.104	0.381	0.713	0.108	0.074	0.575	0.702	0.139	0.092	0.521
PFedRec	0.699	0.145	0.077	0.430	0.722	0.100	0.054	0.530	0.671	0.150	0.078	0.500
FedCD_per	0.704	0.118	0.078	0.588	0.716	0.099	0.068	0.592	0.722	0.101	0.067	0.611
FedCD(FedAvg)	0.710	0.101	0.069	0.613	**0.726**	0.087	0.061	0.608	0.735	0.082	0.056	0.641
FedCD	**0.720**	**0.084**	**0.058**	**0.634**	0.722	**0.072**	**0.053**	**0.622**	**0.737**	**0.068**	**0.048**	**0.658**

5.4.2 Quantitative Fairness Evaluation. The above client-wise comparisons provide intuitive evidence that FedCD can improve the diagnosis performance of underperforming clients, but they do not directly quantify the overall fairness among clients. Therefore, we further conduct a quantitative fairness evaluation using the explicit fairness metrics defined in Section 5.1.3. Conducting the same evaluation across ASSIST2009, ASSIST2012, and SLP-Math is also meaningful because the three datasets exhibit clear yet distinct patterns of client-level response-distribution and data-quantity heterogeneity, as characterized in Table 1. Specifically, we evaluate Avg. ACC, GF, STD_{ACC}, and WACC for different federated approaches under the NCD base model with the 80%/20% train/test split. Here, Avg. ACC is reported to assess overall diagnostic utility, while GF, STD_{ACC}, and WACC are used to evaluate client fairness.

Table 9 compares FedAvg, FedAtt, LoAdaBoost, F2MF, PFedRec, *FedCD_per*, *FedCD(FedAvg)*, and FedCD on ASSIST2009, ASSIST2012, and SLP-Math using NCD under the 80%/20% splitting setting. Here, Avg. ACC denotes the average of client-level ACC values. *FedCD_per* keeps the exercise embeddings locally personalized without

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cross-client aggregation, while *FedCD(FedAvg)* globally shares the exercise embeddings using average aggregation. Therefore, these two variants further help analyze the effects of globally shared exercise embeddings and fairness-aware aggregation.

As shown in Table 9, FedCD achieves the best overall client-fairness performance across the three datasets. Specifically, it obtains the lowest GF and STD_{ACC} and the highest WACC on ASSIST2009, ASSIST2012, and SLP-Math. For example, compared with FedAvg, FedCD reduces GF from 0.167 to 0.084, from 0.121 to 0.072, and from 0.146 to 0.068 on the three datasets, respectively. Meanwhile, STD_{ACC} decreases from 0.112 to 0.058, from 0.081 to 0.053, and from 0.097 to 0.048, while WACC increases from 0.355 to 0.634, from 0.550 to 0.622, and from 0.506 to 0.658. These results consistently indicate smaller group-level disparity, lower overall client-performance dispersion, and better worst-client performance. Importantly, these fairness improvements are accompanied by competitive or better Avg. ACC, suggesting that FedCD improves the balance of client performance without sacrificing overall diagnostic utility.

The comparisons among the three FedCD variants provide further insight into the source of the fairness improvement. Compared with *FedCD_per*, *FedCD(FedAvg)* consistently obtains lower GF and STD_{ACC} and higher WACC, indicating that globally sharing exercise embeddings enables poorly performing clients to benefit from cross-client exercise-related knowledge. FedCD further improves these fairness metrics over *FedCD(FedAvg)*, suggesting that the fairness-aware aggregation mechanism further improves the construction of globally shared exercise representations by giving greater attention to poorly performing clients. Overall, these results provide consistent evidence that FedCD can reduce client-level performance disparity while maintaining promising diagnostic accuracy.

The quantitative fairness results above are obtained with NCD and do not establish the same fairness gains for other CDMs. Local personalization and loss-based aggregation may also help balance client performance in other models, but their effects can depend on the model architecture. For KaNCD, the client-wise ACC results in Table 8 provide encouraging evidence of diagnostic utility, but do not establish overall client fairness. Its more strongly coupled student–exercise–concept representations may also affect the benefit of sharing exercise embeddings, as discussed in RQ1. Evaluating GF, STD_{ACC}, and WACC under KaNCD and other CDMs is therefore needed to assess fairness generalizability.

5.5 Sensitivity of Key Hyperparameters (RQ4)

5.5.1 Sensitivity Analysis of Coefficient. As highlighted, an important hyperparameter γ in Eq.8 represents client importance. To investigate its impact on the performance of the proposed FedCD, we conducted a sensitivity analysis on the ASSIST2009 and SLP-Math datasets. Specifically, we fixed all other parameters in FedCD and varied γ across a range of values from 0.1 to 0.6, with increments of 0.1.

As illustrated in Fig. 5, the proposed FedCD demonstrated different performance on the two datasets due to the differing data distributions among clients in ASSIST2009 and SLP-Math. Notably, the best results were achieved when γ was set to 0.3 for ASSIST2009 and 0.5 for SLP-Math, respectively. However, when the coefficient became too large, the model tended to favor better-performing clients while neglecting those with poorer performance, leading to a decline in overall effectiveness. To strike a balance across the three datasets, we adopted $\gamma = 0.1$, which ensures promising performance across all datasets.

5.5.2 The Influence of Personalization Ratio of Diagnostic Function Parameters . In FedCD, all parameters of the diagnostic function are personalized. To investigate the impact of varying degrees of parameter personalization on model performance, we defined a personalized-to-global parameter ratio and conducted experiments using FedCD based on the NCD model on the ASSIST2009 dataset. With NCD, three variants of FedCD were created: FedCD(0%) with all parameters globally shared, FedCD(w/o W_fc) with parameters of three-layer FC globally shared, and FedCD(w/o W_disc) with parameters of W_disc globally shared.

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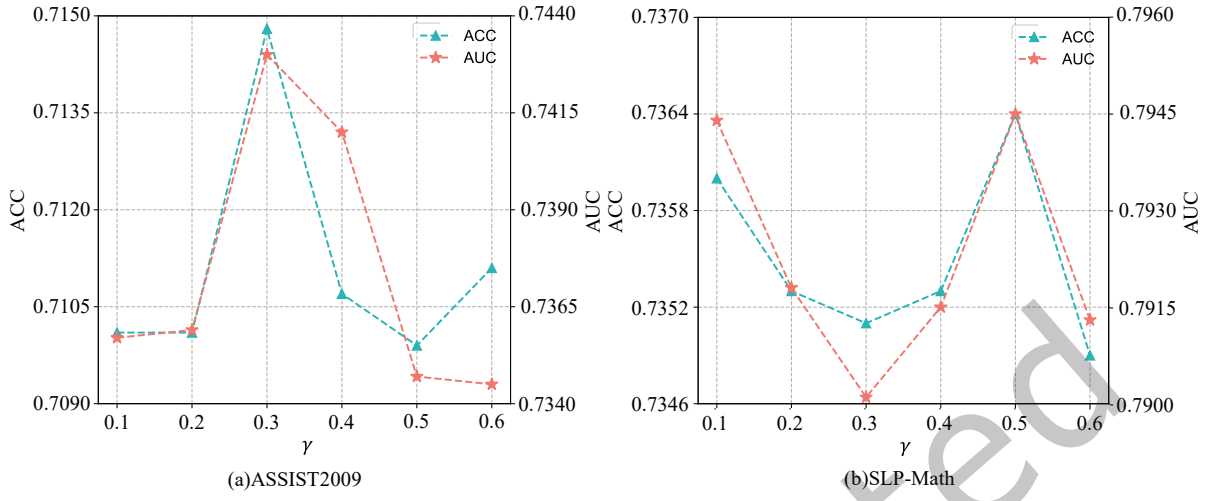


Fig. 5. Sensitivity analysis of FedCD to coefficient γ on the ASSIST2009 and SLP-Math datasets.

Table 10. Different personalization ratio of FedCD on the ASSIST2009 dataset under the NCD base model

Models	ACC↑	RMSE↓	AUC↑
FedCD(0%)	0.7114	0.4587	0.7221
FedCD(w/o W_fc)	0.7151	0.456	0.7383
FedCD(w/o W_disc)	0.7184	0.4453	0.7389
FedCD	**0.7201**	**0.4439**	**0.7413**

Obviously, FedCD holds better performance with more personalized parameters, which means that more parameters can better learn the personalized features of local data.

The results in Table 10 reveal that increasing the degree of personalization consistently improved model performance across all metrics. Full personalization in FedCD yielded the best results, achieving the best performance. This demonstrated that selectively personalizing interaction parameters significantly enhanced the model's ability to adapt to heterogeneous client data, validating the effectiveness of personalized parameter design in FedCD.

5.5.3 Sensitivity Analysis of Local Epoch. In federated learning, the number of local epochs plays a critical role in determining client performance. A smaller number of local epochs often results in undertraining, thereby degrading overall performance, while a larger number increased training time and posed a risk of overfitting. To identify the optimal setting, we conducted experiments on the ASSIST2009 dataset by varying the number of local epochs in FedCD from 1 to 5.

The results in Table 11 show that FedCD maintains stable performance across different local training epochs, with the best ACC achieved at local_epoch 5. This indicates that moderate increases in local updates can enhance model performance without overfitting.

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Table 11. ACC performance of FedCD with various local epoch on ASSIST2009

Local_epoch / ACC↑	**1**	**2**	**3**	**4**	**5**
FedCD	0.7017	0.7142	0.7077	0.7039	0.7201

5.6 Robustness under Privacy-preserving Perturbation (RQ5)

To evaluate the compatibility of FedCD with the adopted privacy-preserving perturbation mechanism, we examine its diagnostic utility under different Laplace-noise intensities. Specifically, the experiment is conducted on ASSIST2009 using NCD as the base model, with δ set to 0.10, 0.20, 0.30, 0.40, and 0.50. Table 12 summarizes the results.

Table 12. Diagnosis performance of FedCD under different Laplace-noise perturbation intensities δ on ASSIST2009

Intensity δ	**0.10**	**0.20**	**0.30**	**0.40**	**0.50**
ACC↑	70.89	70.96	70.13	69.53	68.11
RMSE↓	46.18	46.11	46.07	46.15	46.07
AUC↑	72.96	71.22	70.76	70.62	69.47

As shown in Table 12, ACC and AUC generally decrease as the perturbation intensity increases, while RMSE remains relatively stable. The performance degradation becomes more apparent when δ exceeds 0.30. These results indicate that FedCD retains relatively stable diagnostic utility under moderate perturbation, while excessive noise can adversely affect prediction performance. Therefore, the experiment mainly demonstrates the compatibility and utility robustness of FedCD when combined with the adopted privacy-preserving perturbation mechanism.

5.7 Further Discussion of FedCD (RQ6)

5.7.1 Consistency Evaluation. Intuitively, if student a a demonstrates a higher level of mastery over knowledge concept k compared to student b, then student a is expected to have a greater likelihood of correctly answering exercises that involve concept k. To evaluate the ranking performance with respect to such relative proficiency, we adopted the Degree of Agreement (DOA) as the assessment metric. For a given knowledge concept k, the DOA score, denoted as DOA(k), is defined as follows:

$$\mathrm{DOA}(k) = \frac{1}{Z}\sum_{a=1}^{N}\sum_{b=1}^{N}\delta\left(F_{ak}^{s}, F_{bk}^{s}\right)\sum_{j=1}^{M} I_{jk}\frac{J(j,a,b)\wedge\delta\left(r_{aj}, r_{bj}\right)}{J(j,a,b)}, \tag{17}$$

let $Z = \sum_{a=1}^{N}\sum_{b=1}^{N}\delta\left(F_{ak}^{s}, F_{bk}^{s}\right)$, where F_{ak}^{s} represents the estimated proficiency of student a on concept k, and $\delta(x,y) = 1$ if x>y and $\delta(x,y) = 0$ otherwise. The indicator $I_{jk} = 1$ if exercise j is associated with knowledge concept k, and $I_{jk} = 0$ otherwise. Similarly, $J(j,a,b) = 1$ if both student a and b have attempted exercise j, and $J(j,a,b) = 0$ otherwise. The average of DOA(k) across all knowledge concepts is employed to measure the overall quality of the diagnostic outcomes, i.e., the accuracy of the knowledge proficiency estimates provided by the model. The experiment was conducted on ASSIST2009 with NCD-based FedAvg, FedAtt, FedRAP, and FedCD.

As shown in Table 13, FedCD achieves the highest DOA score, demonstrating superior capability in capturing students' relative proficiencies. This is mainly due to its personalized FL design, which tailors local models to

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Table 13. DOA and ACC values of FedCD, FedAvg, FedAtt, and FedRAP on the ASSIST2009 dataset under the NCD base model

Models	DOA↑	ACC↑
FedAvg	0.557	0.680
FedAtt	0.560	0.675
FedRAP	0.602	0.701
FedCD	**0.612**	**0.720**

individual student distributions. Unlike global-only approaches, FedCD better preserves proficiency rankings by accounting for learner diversity, leading to improved interpretability and diagnostic quality.

5.7.2 The Influence of Varying the Loss Function Weights. In FedCD, to promote fairness, we assigned equal loss weights (set to 1) to all clients, encouraging the model to give equal attention to clients with smaller data volumes. This approach contrasts with previous methods such as FedAvg, which prioritizes clients with larger data volumes and often neglects those with fewer samples. To evaluate the effectiveness of this fairness-enhancing strategy, we modified the aggregation of client losses in FedCD by weighting each client's loss according to its proportion of the total data volume. We denoted this variant as FedCD_loss_avg, which differs from the FedCD (FedAvg) in our paper, where the parameter aggregation weights are based on data volume. To evaluate the impact, we randomly selected one-fourth of the schools and presented the comparison results. Both FedCD and FedCD_loss_avg were implemented based on the NCD model and evaluated on the ASSIST2009 dataset.

Table 14. AUC performance comparison on schools Between FedCD and FedCD_loss_avg on the ASSIST2009 dataset

School Id	Log_num	AUC↑	
		FedCD_loss_avg	FedCD
2	99306	0.7392	0.7337
3	9766	0.6666	0.6720
5	11463	0.7613	0.7545
9	1085	0.6377	0.6397
12	4173	0.7392	0.7333

These results suggest that modifying the loss aggregation weights provides an effective means to promote fairness in federated cognitive diagnosis. As shown in Table 14, FedCD_loss_avg and FedCD both perform well on Schools 2, 5, and 12, but since FedCD_loss_avg inherently favors clients with larger data volumes, its performance on these schools is even better. Conversely, on Schools 3 and 9, where both methods already perform poorly, FedCD_loss_avg performs even worse due to underweighting clients with fewer samples.

5.7.3 Statistics of Computational Overhead. For FedCD, its each round (P rounds) of computational overhead includes training each client (T clients) of models for Num epochs and aggregating exercise parameters (with dimensions $M \cdot D$). $|R|$ denotes training data size. The first part is about $O(T \cdot Num \cdot |R| \cdot \hat{d})$, where $O(\hat{d})$ denotes overhead of one forward and one backward pass; the second is about $O(T \cdot (1 + M \cdot D))$. Totally, FedCD holds $O(P \cdot (T \cdot Num \cdot |R| \cdot \hat{d} + T \cdot (1 + M \cdot D)))$. Compared to other methods, the different overhead lies in the second

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part. Although FedCD's second part may be worse than others, it has much less impact on the overhead than the first part.

Similarly, the communication overhead of FedCD is $O(2 \cdot M \cdot D)$, which is lower than that of non-personalized methods such as FedAvg, as it does not require transmitting the entire set of model parameters.

To evaluate the practical time overhead of different FL methods, we selected two representative approaches, FedAvg and FedCD, and conducted experiments on the ASSIST2009 dataset.

Table 15. Time consumption of FedCD and FedAvg on the ASSIST2009 dataset under the NCD base model

Seconds↓	**FedAvg**	**FedCD**
Train_time	579.76	588.18
Evaluation_time	102.15	103.94

As shown in Table 15, although FedCD employs a more complex model structure than FedAvg, the increase in training and evaluation time is less than 10 seconds in total. Despite the slight increase in computational cost, FedCD achieves significantly better performance, demonstrating its efficiency and practicality in real-world FL scenarios.

6 Conclusion and Future Work

This paper proposed a fairness-aware federated cognitive diagnosis framework FedCD. The proposed FedCD follows the basic procedures of common FL approaches but adopts a novel parameter decoupling-based personalization strategy. The main idea of this strategy is to divide model parameters into two parts: a locally personalized part and a globally shared part, where the former allows each client to have personalized parameters to learn local data features. For the latter, the fairness-aware parameter aggregation mechanism is used to update the exercise embedding parameters to address the client fairness issue. Compared to five FL approaches across three CDMs on three datasets, the FedCD demonstrates competitive performance, and its capability to address the issue of client fairness has also been validated.

Although the three datasets cover different client settings, they do not provide a controlled evaluation of how FedCD's fairness gains change as client heterogeneity increases. Differences in response distributions, exercise and knowledge-concept coverage, and data volumes may jointly affect both diagnostic performance and client fairness, making their individual effects difficult to distinguish. In future work, we will construct a wider range of client settings by selecting subsets of schools or merging and splitting school clients without altering response labels. We will examine these factors while accounting for changes in client numbers and data volumes, and jointly evaluate diagnostic performance and fairness. Such analyses will help identify the conditions under which fairness-aware exercise sharing is most beneficial and where more adaptive personalization is needed.

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References

[1] Khan Academy. 2024. Khan Academy. https://www.khanacademy.org/

[2] Mohammed Adnan, Shivam Kalra, Jesse C Cresswell, Graham W Taylor, and Hamid R Tizhoosh. 2022. Federated learning and differential privacy for medical image analysis. *Scientific reports* 12, 1 (2022), 1953.
[3] Md Zahangir Alom, Tarek M Taha, Christopher Yakopcic, Stefan Westberg, Paheding Sidike, Mst Shamima Nasrin, Brian C Van Esesn, Abdul A S Awwal, and Vijayan K Asari. 2018. The history began from alexnet: A comprehensive survey on deep learning approaches. *arXiv preprint arXiv:1803.01164* (2018).
[4] Di Chai, Leye Wang, Kai Chen, and Qiang Yang. 2020. Secure federated matrix factorization. *IEEE Intelligent Systems* 36, 5 (2020), 11–20.
[5] Coursera. 2024. Coursera. https://www.coursera.org/
[6] Jiajun Cui, Hong Qian, Chanjin Zheng, Lu Wang, Mo Yu, and Wei Zhang. 2025. Rebalancing Discriminative Responses for Knowledge Tracing. *ACM Transactions on Information Systems* (2025).
[7] Joaquín Derrac, Salvador García, Daniel Molina, and Francisco Herrera. 2011. A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. *Swarm and Evolutionary Computation* 1, 1 (2011), 3–18.
[8] John C. Duchi, Michael I. Jordan, and Martin J. Wainwright. 2013. Local Privacy and Statistical Minimax Rates. In *2013 IEEE 54th Annual Symposium on Foundations of Computer Science*. IEEE, 429–438.
[9] Susan E Embretson and Steven P Reise. 2013. *Item response theory*. Psychology Press.
[10] Mingyu Feng, Neil Heffernan, and Kenneth Koedinger. 2009. Addressing the assessment challenge with an online system that tutors as it assesses. *User modeling and user-adapted interaction* 19 (2009), 243–266.
[11] Weibo Gao, Qi Liu, Zhenya Huang, Yu Yin, Haoyang Bi, Mu-Chun Wang, Jianhui Ma, Shijin Wang, and Yu Su. 2021. Rcd: Relation map driven cognitive diagnosis for intelligent education systems. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 501–510.
[12] Yang Gao, Hong Yang, Peng Zhang, Chuan Zhou, and Yue Hu. 2021. Graph neural architecture search. In *Proceedings of the Twenty-Ninth International Conference on International Joint Conferences on Artificial Intelligence*. 1403–1409.
[13] Tural Gurbanov, Francesco Ricci, and Meinhard Ploner. 2016. Modeling and predicting user actions in recommender systems. In *Proceedings of the 2016 Conference on User Modeling Adaptation and Personalization*. 151–155.
[14] Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. 2017. Neural collaborative filtering. In *Proceedings of the 26th international conference on world wide web*. 173–182.
[15] Sepp Hochreiter. 1998. The vanishing gradient problem during learning recurrent neural nets and problem solutions. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems* 6, 02 (1998), 107–116.
[16] Liya Hu, Zhiang Dong, Jingyuan Chen, Guifeng Wang, Zhihua Wang, Zhou Zhao, and Fei Wu. 2023. PTADisc: A Cross-Course Dataset Supporting Personalized Learning in Cold-Start Scenarios. In *Proceedings of the 37th Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
[17] Li Huang, Yifeng Yin, Zeng Fu, Shifa Zhang, Hao Deng, and Dianbo Liu. 2020. LoAdaBoost: Loss-based AdaBoost federated machine learning with reduced computational complexity on IID and non-IID intensive care data. *Plos one* 15, 4 (2020), e0230706.
[18] Zhenya Huang, Qi Liu, Yuying Chen, Le Wu, Keli Xiao, Enhong Chen, Haiping Ma, and Guoping Hu. 2020. Learning or forgetting? A dynamic approach for tracking the knowledge proficiency of students. *ACM Transactions on Information Systems (TOIS)* 38, 2 (2020), 1–33.
[19] Frank Hutter, Lars Kotthoff, and Joaquin Vanschoren. 2019. *Automated machine learning: methods, systems, challenges*. Springer Nature.
[20] Shaoxiong Ji, Shirui Pan, Guodong Long, Xue Li, Jing Jiang, and Zi Huang. 2019. Learning private neural language modeling with attentive aggregation. In *2019 International joint conference on neural networks (IJCNN)*. IEEE, 1–8.
[21] Yaochu Jin, Hangyu Zhu, Jinjin Xu, and Yang Chen. 2023. *Federated Learning*. Springer.
[22] Peter Kairouz, H Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, et al. 2021. Advances and open problems in federated learning. *Foundations and trends® in machine learning* 14, 1–2 (2021), 1–210.
[23] Yehuda Koren, Robert Bell, and Chris Volinsky. 2009. Matrix factorization techniques for recommender systems. *Computer* 42, 8 (2009), 30–37.
[24] Jiatong Li, Fei Wang, Qi Liu, Mengxiao Zhu, Wei Huang, Zhenya Huang, Enhong Chen, Yu Su, and Shijin Wang. 2022. Hiercdf: A bayesian network-based hierarchical cognitive diagnosis framework. In *Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining*. 904–913.
[25] Li Li, Yuxi Fan, Mike Tse, and Kuo-Yi Lin. 2020. A review of applications in federated learning. *Computers & Industrial Engineering* 149 (2020), 106854.
[26] Qingyao Li, Wei Xia, Li'ang Yin, Jian Shen, Renting Rui, Weinan Zhang, Xianyu Chen, Ruiming Tang, and Yong Yu. 2023. Graph enhanced hierarchical reinforcement learning for goal-oriented learning path recommendation. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*. 1318–1327.
[27] Xiang Li, Kaixuan Huang, Wenhao Yang, Shusen Wang, and Zhihua Zhang. 2019. On the convergence of fedavg on non-iid data. *arXiv preprint arXiv:1907.02189* (2019).

Just Accepted

[28] Yujie Lin, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Dongxiao Yu, Jun Ma, Maarten de Rijke, and Xiuzhen Cheng. 2020. Meta matrix factorization for federated rating predictions. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 981–990.

[29] Fei Liu, Chenyang Bu, Haotian Zhang, Le Wu, Kui Yu, and Xuegang Hu. 2024. FDKT: Towards an interpretable deep knowledge tracing via fuzzy reasoning. *ACM Transactions on Information Systems* 42, 5 (2024), 1–26.

[30] Qi Liu, Jinze Wu, Zhenya Huang, Hao Wang, Yuting Ning, Ming Chen, Enhong Chen, Jinfeng Yi, and Bowen Zhou. 2023. Federated user modeling from hierarchical information. *ACM Transactions on Information Systems* 41, 2 (2023), 1–33.

[31] Shuchang Liu, Yingqiang Ge, Shuyuan Xu, Yongfeng Zhang, and Amelie Marian. 2022. Fairness-aware Federated Matrix Factorization. In *Proceedings of the 16th ACM Conference on Recommender Systems* (Seattle, WA, USA) *(RecSys '22)*. Association for Computing Machinery, New York, NY, USA, 168–178. https://doi.org/10.1145/3523227.3546771

[32] Shuo Liu, Junhao Shen, Hong Qian, and Aimin Zhou. 2024. Inductive Cognitive Diagnosis for Fast Student Learning in Web-Based Intelligent Education Systems. In *Proceedings of the ACM on Web Conference 2024*. 4260–4271.

[33] Haiping Ma, Manwei Li, Le Wu, Haifeng Zhang, Yunbo Cao, Xingyi Zhang, and Xuemin Zhao. 2022. Knowledge-Sensed Cognitive Diagnosis for Intelligent Education Platforms. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 1451–1460.

[34] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*. PMLR, 1273–1282.

[35] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*. PMLR, 1273–1282.

[36] Dinh C Nguyen, Ming Ding, Pubudu N Pathirana, Aruna Seneviratne, Jun Li, and H Vincent Poor. 2021. Federated learning for internet of things: A comprehensive survey. *IEEE Communications Surveys & Tutorials* 23, 3 (2021), 1622–1658.

[37] Georgina Peake and Jun Wang. 2018. Explanation mining: Post hoc interpretability of latent factor models for recommendation systems. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining*. 2060–2069.

[38] Vasileios Perifanis and Pavlos S Efraimidis. 2022. Federated neural collaborative filtering. *Knowledge-Based Systems* 242 (2022), 108441.

[39] Yanjun Pu, Fang Liu, Rongye Shi, Haitao Yuan, Ruibo Chen, Tianhao Peng, and Wenjun Wu. 2024. ELAKT: Enhancing locality for attentive knowledge tracing. *ACM Transactions on Information Systems* 42, 4 (2024), 1–27.

[40] Zhuang Qi, Lei Meng, Zhaochuan Li, Han Hu, and Xiangxu Meng. 2025. Cross-silo feature space alignment for federated learning on clients with imbalanced data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 19986–19994.

[41] Mark D Reckase. 2009. Multidimensional item response theory models. In *Multidimensional item response theory*. Springer, 79–112.

[42] Huma Shafiq, Zahid Ashraf Wani, Iram Mukhtar Mahajan, and Uzma Qadri. 2017. Courses beyond borders: A case study of MOOC platform Coursera. *Library Philosophy and Practice* (2017), 1–15.

[43] Xian Shuai, Yulin Shen, Siyang Jiang, Zhihe Zhao, Zhenyu Yan, and Guoliang Xing. 2022. BalanceFL: Addressing class imbalance in long-tail federated learning. In *2022 21st ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN)*. IEEE, 271–284.

[44] Yanan Sun, Bing Xue, Mengjie Zhang, and Gary G Yen. 2019. Evolving deep convolutional neural networks for image classification. *IEEE Transactions on Evolutionary Computation* 24, 2 (2019), 394–407.

[45] D. L. Torre and J. 2009. DINA Model and Parameter Estimation: A Didactic. *Journal of Educational and Behavioral Statistics* 34, 1 (2009), 115–130.

[46] Tung Vuong, Pritom Kumar Das, and Tuukka Ruotsalo. 2025. Incorporating Cognitive Abilities into Web Search Re-ranking. *ACM Transactions on Information Systems* (May 2025). doi:10.1145/3736401 Just Accepted.

[47] Fei Wang, Weibo Gao, Qi Liu, Jiatong Li, Guanhao Zhao, Zheng Zhang, Zhenya Huang, Mengxiao Zhu, Shijin Wang, Wei Tong, et al. 2024. A Survey of Models for Cognitive Diagnosis: New Developments and Future Directions. *arXiv preprint arXiv:2407.05458* (2024).

[48] Fei Wang, Qi Liu, Enhong Chen, Zhenya Huang, Yuying Chen, Yu Yin, Zai Huang, and Shijin Wang. 2020. Neural cognitive diagnosis for intelligent education systems. In *Proceedings of the 2020 AAAI Conference on Artificial Intelligence*, Vol. 34. 6153–6161.

[49] Fei Wang, Qi Liu, Enhong Chen, Zhenya Huang, Yu Yin, Shijin Wang, and Yu Su. 2022. NeuralCD: a general framework for cognitive diagnosis. *IEEE Transactions on Knowledge and Data Engineering* (2022).

[50] Lixu Wang, Shichao Xu, Xiao Wang, and Qi Zhu. 2021. Addressing class imbalance in federated learning. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 35. 10165–10173.

[51] Shanshan Wang, Zhen Zeng, Xun Yang, and Xingyi Zhang. 2023. Self-supervised Graph Learning for Long-tailed Cognitive Diagnosis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 110–118.

[52] Xinping Wang, Caidie Huang, Jinfang Cai, and Liangyu Chen. 2021. Using knowledge concept aggregation towards accurate cognitive diagnosis. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*. 2010–2019.

[53] Chuhan Wu, Fangzhao Wu, Lingjuan Lyu, Tao Qi, Yongfeng Huang, and Xing Xie. 2022. A federated graph neural network framework for privacy-preserving personalization. *Nature Communications* 13, 1 (2022), 3091.

Just Accepted

[54] Jinze Wu, Zhenya Huang, Qi Liu, Defu Lian, Hao Wang, Enhong Chen, Haiping Ma, and Shijin Wang. 2021. Federated deep knowledge tracing. In *Proceedings of the 14th ACM international conference on web search and data mining*. 662–670.
[55] Jinze Wu, Qi Liu, Zhenya Huang, Yuting Ning, Hao Wang, Enhong Chen, Jinfeng Yi, and Bowen Zhou. 2021. Hierarchical personalized federated learning for user modeling. In *Proceedings of the Web Conference 2021*. 957–968.
[56] Zhengyang Wu, Ming Li, Yong Tang, and Qingyu Liang. 2020. Exercise recommendation based on knowledge concept prediction. *Knowledge-Based Systems* 210 (2020), 106481.
[57] Liu Yang, Ben Tan, Vincent W Zheng, Kai Chen, and Qiang Yang. 2020. Federated recommendation systems. *Federated Learning: Privacy and Incentive* (2020), 225–239.
[58] Shangshang Yang, Haiping Ma, Ying Bi, Ye Tian, Limiao Zhang, Yaochu Jin, and Xingyi Zhang. 2024. An Evolutionary Multi-Objective Neural Architecture Search Approach to Advancing Cognitive Diagnosis in Intelligent Education. *IEEE Transactions on Evolutionary Computation* (2024).
[59] Shangshang Yang, Haiping Ma, Cheng Zhen, Ye Tian, Limiao Zhang, Yaochu Jin, and Xingyi Zhang. 2023. Designing novel cognitive diagnosis models via evolutionary multi-objective neural architecture search. *arXiv preprint arXiv:2307.04429* (2023).
[60] Shangshang Yang, Ye Tian, Xiaoshu Xiang, Shichen Peng, and Xingyi Zhang. 2022. Accelerating Evolutionary Neural Architecture Search via Multifidelity Evaluation. *IEEE Transactions on Cognitive and Developmental Systems* 14, 4 (2022), 1778–1792.
[61] Shangshang Yang, Haoyu Wei, Haiping Ma, Ye Tian, Xingyi Zhang, Yunbo Cao, and Yaochu Jin. 2023. Cognitive diagnosis-based personalized exercise group assembly via a multi-objective evolutionary algorithm. *IEEE Transactions on Emerging Topics in Computational Intelligence* (2023).
[62] Shangshang Yang, Cheng Zhen, Ye Tian, Haiping Ma, Yuanchao Liu, Panpan Zhang, and Xingyi Zhang. 2023. Evolutionary Multi-Objective Neural Architecture Search for Generalized Cognitive Diagnosis Models. In *Proceedings of the 5th International Conference on Data-driven Optimization of Complex Systems*. IEEE, 1–10.
[63] Shangshang Yang, Cheng Zhen, Ye Tian, Haiping Ma, Yuanchao Liu, Panpan Zhang, and Xingyi Zhang. 2023. Evolutionary Multi-Objective Neural Architecture Search for Generalized Cognitive Diagnosis Models. In *2023 5th International Conference on Data-driven Optimization of Complex Systems (DOCS)*. IEEE, 1–10.
[64] Timothy Yang, Galen Andrew, Hubert Eichner, Haicheng Sun, Wei Li, Nicholas Kong, Daniel Ramage, and Françoise Beaufays. 2018. Applied federated learning: Improving google keyboard query suggestions. *arXiv preprint arXiv:1812.02903* (2018).
[65] Wensi Yang, Yuhang Zhang, Kejiang Ye, Li Li, and Cheng-Zhong Xu. 2019. Ffd: A federated learning based method for credit card fraud detection. In *Big Data–BigData 2019: 8th International Congress, Held as Part of the Services Conference Federation, SCF 2019, San Diego, CA, USA, June 25–30, 2019, Proceedings 8*. Springer, 18–32.
[66] Han Yu, Zelei Liu, Yang Liu, Tianjian Chen, Mingshu Cong, Xi Weng, Dusit Niyato, and Qiang Yang. 2020. A fairness-aware incentive scheme for federated learning. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 393–399.
[67] Chunxu Zhang, Guodong Long, Tianyi Zhou, Peng Yan, Zijian Zhang, Chengqi Zhang, and Bo Yang. 2023. Dual personalization on federated recommendation. *arXiv preprint arXiv:2301.08143* (2023).
[68] Chunxu Zhang, Guodong Long, Tianyi Zhou, Zijian Zhang, Peng Yan, and Bo Yang. 2024. GPFedRec: Graph-Guided Personalization for Federated Recommendation. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 4131–4142.
[69] Chunxu Zhang, Guodong Long, Tianyi Zhou, Zijian Zhang, Peng Yan, and Bo Yang. 2024. When Federated Recommendation Meets Cold-Start Problem: Separating Item Attributes and User Interactions. In *Proceedings of the ACM Web Conference 2024* (Singapore, Singapore) *(WWW '24)*. Association for Computing Machinery, New York, NY, USA, 3632–3642. doi:10.1145/3589334.3645525
[70] Chen Zhang, Yu Xie, Hang Bai, Bin Yu, Weihong Li, and Yuan Gao. 2021. A survey on federated learning. *Knowledge-Based Systems* 216 (2021), 106775.
[71] Zheng Zhang, Wei Song, Qi Liu, Qingyang Mao, Yiyan Wang, Weibo Gao, Zhenya Huang, Shijin Wang, and Enhong Chen. 2024. Towards accurate and fair cognitive diagnosis via monotonic data augmentation. *Advances in Neural Information Processing Systems* 37 (2024), 47767–47789.
[72] Hangyu Zhu, Jinjin Xu, Shiqing Liu, and Yaochu Jin. 2021. Federated learning on non-IID data: A survey. *Neurocomputing* 465 (2021), 371–390.
[73] Barret Zoph and Quoc V Le. 2016. Neural architecture search with reinforcement learning. *arXiv preprint arXiv:1611.01578* (2016).

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